Design and Implementation of Dual Active Bridge Converter

A

Thesis

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by

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Abstract

Worldwide, with growing awareness regarding environmental impacts such as global warming and pollution, the trend of adopting electric vehicles is becoming more prevalent than their traditional counterparts. This project investigates the growth rate of the adoption of EVs in India through an assessment of the market dynamics, manufacturers, and distribution of EVs across various classes of vehicles, including two-wheelers, four-wheelers, trucks, and buses. It also considers the common voltage and power rating ranges utilized in EVs. Conversion and regulation of power is necessary in the case of charging EV batteries. Since the source power available for use in the EVs is in the form of AC power, its conversion and regulation are necessary. This conversion and regulation process is usually implemented in multiple stages. The first stage involves the conversion of AC power to DC power, and then there is the conversion from DC to DC power. In this study, a system consisting of two stages: AC to DC converter with boost stage and DAB converter, will be considered. Power factor correction using unit vector template control technique will be implemented. The DAB converter is used as an isolated DC-DC stage, which provides galvanic isolation and allows efficient power transfer at high frequency. Due to its bidirectional capability and reduced switching losses under soft-switching conditions, it is suitable for EV charging applications. A hardware setup of the DAB converter is developed, where the gate driver circuit is operated using an FPGA. The overall work focuses on understanding the system operation through simulation and validating it through hardware implementation.

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Chapter 1

Introduction

With the increasing demand for sustainable energy and the environmental impact of conventional fuels, electric vehicles (EVs) are gaining significant importance worldwide. In the simplest sense, electric vehicles employ electric motors to propel the vehicle rather than the conventional internal combustion engine. The major components of an electric vehicle are the battery, a motor used to drive the EV, a controller for the motor, a battery management system (BMS), and power electronics converters [1]. The generalized block diagram for EVs is shown in Fig. 1.1. In India, the growth of EVs has accelerated in recent years due to government initiatives, policy support, and advancements in power electronics and battery technologies. As EV adoption increases, the need for efficient and reliable charging infrastructure becomes equally important. This creates new challenges in terms of power quality, efficiency, and compatibility with the existing electrical grid.

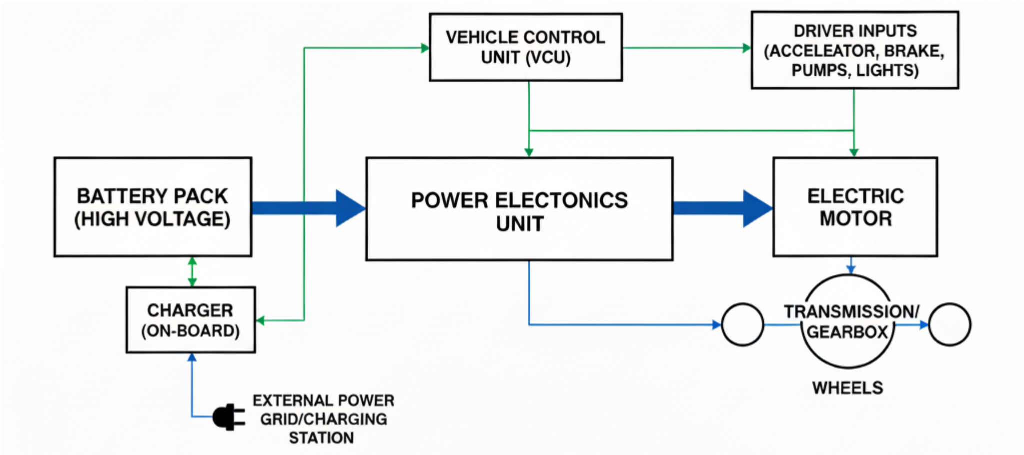


Figure 1.1: Electric Vehicle System Block Diagram

An important issue with most EV charging stations is poor quality power drawn from the grid. Most chargers utilize non-sinusoidal currents, and therefore, the power factor and harmonic distortions are low. As a result, this leads to poor efficiency and possibly even impacts the integrity of the grid power distribution. To improve these characteristics, Power Factor Correction is implemented into the charger system so that the drawn input current is closer to the input voltage.

Since there are many risks in connecting the charging system to the grid, isolated DC-DC converters are essential in transferring energy from the source to the battery safely. In particular, the Dual Active Bridge (DAB) converter became the preferred solution due to the benefits it offers. First of all, the DAB converter is based on two full-bridge circuits coupled by a high-frequency transformer that provides isolation and greater flexibility in operations. High frequencies allow reducing the sizes of passive elements and therefore increasing the efficiency and reducing the size of the system. Another important feature of the DAB converter is its ability to control the power transfer through the phase shift between the input and output bridge circuits. At the same time, one of the advantages is also soft switching operation when it comes to Zero Voltage Switching (ZVS). The last factor makes it applicable for EV charging systems and helps in achieving higher efficiency.

However, apart from choosing the appropriate solution, implementation is also crucial for efficient performance. Field Programmable Gate Array devices have been widely used in modern power electronics since they provide parallelism and fast operations required for high-frequency switching. FPGAs allow receiving the precise switching signal and therefore increase the efficiency. Thus, FPGA is a good option for DAB conversion implementation.

The purpose of the work is to study the development of EVs in India and implement an EV charging system with PFC and DAB converter. Specifically, the goal is to implement hardware realization of DAB converter using the FPGA platform. Simulation of the converter will be carried out to study its operation and performance parameters.

1.1 Literature Survey

Electric Vehicles (EVs) are currently experiencing rapid adoption within the transportation sector [2]. They can also be used to couple renewable energy sources to provide energy management in smart grids [3] [4]. The charging process involves converting AC voltage to a specific DC level which can be achieved via active and passive converters [5].

In recent times, there has been a great increase in the number of electric vehicles (EVs) with European automakers making commitments to have all their cars fully electric in the coming few decades. However, despite their growing popularity, there are certain barriers hindering the adoption of electric battery cars such as high prices, lack of range, longer charging periods, and inadequate infrastructure [6].

India is not left out of the trend as many electric vehicles have been adopted within the country due to its policies which include carbon emission reductions, reduction in dependency on imported fossil fuels, and sustainability considerations. There are several government-led schemes and missions to promote the adoption of EVs such as the National Electric Mobility Mission Plan (NEMMP) and Faster Adoption and Manufacturing of Electric Vehicles (FAME) scheme [7].

Fast charging of EVs has also been the focus of much research and development. Fast charging and ultra-fast charging is vital for enhancing user experience, especially during intercity traveling. Recent advances in this field include charging that aims at delivering up to 80% battery charge within a very short time using extremely high-power systems reaching 350 kW in some cases. However, this is limited by other battery factors including SoC, temperatures, and battery chemistry [6].

Many researchers have investigated the best techniques for the fast charging of EV batteries. This includes studies on the performance of the Constant Current–Constant Voltage (CCCV) charging process. Recent developments include advanced battery charging methodologies including pulsing techniques including Positive Pulse Charging (PPC) and Sinusoidal Ripple Charging (SRC). Such techniques help to improve the overall performance of the battery in terms of lifespan and temperature rise among others. However, they also have disadvantages such as increased complexity [8].

There are various ways in which power electronics helps in efficient energy management and coupling renewable energy sources to the grid. As a result of rising penetration of renewables, advanced converter topologies are needed for reliability and efficiency improvements. One of the major improvements is associated with the use of wide bandgap semiconductors such as SiC and GaN which offer improved efficiency due to high switching frequencies and reduced losses [9].

Of the available DC-DC converter topologies, Dual Active Bridge (DAB) converters have received considerable attention due to their bidirectional capability, galvanic isolation, efficiency, and compact size. This converter topology consists of two active bridges coupled by a high frequency transformer with power being controlled through regulating the phase shift between the primary and secondary bridges. It finds applicability in battery charging and energy storage [10].

Several investigations regarding the dynamics and steady-state behavior of DAB converters have been done. High efficiency is made possible by Zero Voltage Switching (ZVS) throughout the converter's operating range. Moreover, the ability to operate in both step-up and step-down modes makes it suitable for various applications in battery systems, renewable sources, and DC microgrids [11].

Further improvements in DAB converter performance include modeling and simulation which is important for designing controllers. Various small signal models show superior performance in controlling the dynamic behavior and voltage regulation of the converters when operated under different conditions [12].

In terms of control, SPS control is popular because of its simplicity. However, there are certain drawbacks of this control strategy such as low efficiency and increased reactive power losses. Some of the alternative controls include

DPS control and EPS control with advantages including better efficiency and reduced current stress besides extended soft switching zones [13].

The design of HFT also constitutes one of the most researched areas in DAB converters since it is an important component of the converter. The role of the transformer is to enable galvanic isolation and contribute to power transfer through leakage inductance of the transformer. The correct choice of core material and proper winding configuration is important for good performance of the device [14].

A comprehensive review of DAB converters is provided by [15]. The comparison of different types of models including reduced order, generalized average, and discrete time is done in terms of efficiency. The conclusion reached is that reduced order models have a good trade-off between performance and accuracy whereas discrete time models perform better under fast switching conditions. The paper goes on to discuss several control techniques including feedback, feed forward, model predictive, and sliding mode controls as well as challenges associated with implementations such as dead-time effects, phase drift, and imbalance magnetic flux.

1.2 Market Survey of Electric Vehicles (EVs) in India

1.2.1 Overview of the Indian EV Market

The Indian EV market has been moving at a pace that even policymakers didn't fully expect. Over the last few years, sales have climbed sharply, 2022 was the first year to cross the 10 lakh mark, which was a big milestone. And then, by just the middle of 2023, the country had already reached more than 73% of the previous year's sales. Most of this momentum has come from electric two-wheelers and three-wheelers. They are highly appropriate for travel within cities where there are high population densities, and they provide more convenience for short distance travels on a daily basis. This rise has not come about by coincidence; rather, there have been several contributing factors like government subsidies and low battery prices [7].

Segment Trends: Two-Wheelers, Three-Wheelers, Cars, and Buses

1. **Two-wheelers (e2W):** This segment holds the lead in terms of dominance in the EV industry, making up around 80-85% of total EV sales in India. The preference for these models is due to their affordability, simplicity in operating, and low maintenance needs. They are also favored by the policies of the government. They have made significant contributions to the growth of EVs in India [16].

2. **Three-wheelers (e3W):** This section accounts for around 10-12% of the entire market. It has proved to be a pragmatic solution to traditional auto-rickshaws and delivery vans. The vehicles are deemed cost-efficient to their users, and with rising fuel costs, the trend towards using electric cars has become financially viable. The vehicles are mainly used in urban and semi-urban zones and are mainly employed for transportation of people and goods.

3. **Four-wheelers (electric cars):** While it is true that the four wheeler EV market is still smaller compared to others, this segment has been experiencing rapid growth in recent years. The increasing involvement of manufacturers in addition to the acceptance of consumers and the development of the infrastructure for charging has made the segment experience growth [7].

4. **Electric Buses:** India has become the second-biggest market of electric buses in the world after China. Until 2025, there were more than 9,000 e-buses running across different states in India. This was made possible mainly because of the FAME II subsidy scheme through which the acquisition of e-buses became easier on a large scale. It has been found that there is a lot of activity in major cities such as Delhi, Mumbai, Bengaluru, and Chennai. Long-term operating contracts and large-scale purchases are helping them to scale up operations.

5. **Electric Trucks and Commercial Fleets:** While the industry is at a nascent stage, the sector is slowly picking up pace with the entry of Tata Motors, Ashok Leyland, Eicher, Omega Seiki into the segment by rolling out electric variants of their vehicles meant for urban logistics services. Several issues remain, such as payload capacity, driving range, and lack of charging stations. Nevertheless, several logistic service providers, as well as e-commerce companies, have embarked on pilot projects and deployed some fleets driven by sustainable initiatives [16].

Technical and Policy Barriers

The barriers identified are:

1. **Limited and unevenly distributed charging stations:** Charging infrastructure isn't keeping up with sales. Fast chargers especially tend to cluster around big cities, leaving many regions underserved.

2. **Battery standardization and vehicle interoperability issues:** Different companies follow different systems connectors, voltages, cell formats, and that makes interoperability a real challenge. This slows down scalability.

3. **High upfront cost:** Despite dropping battery prices, several EV categories especially buses and trucks are still significantly more expensive than their ICE counterparts. For many state transport agencies, this becomes a financial hurdle.

4. **Policy fragmentation across states:** Each state has its own policy structure, subsidy model, and implementation speed. This often results in an EV ecosystem that feels a bit uneven and complicated.

5. **Range anxiety and TCO (Total Cost of Ownership) challenges:** Consumers still worry about range, long term running costs, and battery life, regardless of the actual improvements in technology [17].

Forecasts and Stakeholder Impact

By 2030, there will be estimates that electric vehicles could reach a level of up to 30% of the new vehicle sales in the nation. The largest share will come from e2W, e3W, and bus categories. Many studies on technologies and policies have mentioned the important contribution of electric buses and trucks towards reducing air pollution and improving logistical efficiency in urban environments. Nevertheless, this goal cannot be accomplished without proper planning and financial support [18].

Consumer & Demographic Insights

Studies reveal that there is a significant number of individuals who have purchased their electric vehicles due to personal interest in this field. In contrast, the demand for electric buses and other electric vehicles in commercial organizations depends heavily on government tenders, environmental, social, and governance goals of companies, and urban decarbonization policies [19].

Industry, News, and Contemporary Data

By the close of 2024, more than 25,000 charging points were set up in India. The rollout of electric buses through the FAME II program was becoming ever more popular in large urban centers. Simultaneously, private logistics firms and e-commerce operators started showing interest in electric trucks and vans. Their ultimate aim is for a considerable share of their last-mile delivery vehicles to be electric by 2030.

1.2.2 Market Share of EV Companies in India

Market Share of Indian EV Companies

Four-Wheeler Segment **Tata Motors** still sits at the top of India's EV four-wheeler market, although its grip isn't as firm as before. In September 2025, Tata's share stayed above 40%, which is strong, but definitely lower than the peak it enjoyed in 2024 when it held more than 60%. The competition has become more intense since MG Motor and Mahindra have pushed more models and improved availability. So, while Tata remains the leader, it's no longer the only giant in the room.

Mahindra & Mahindra has been making noticeable progress. With their BE 6 and XEV 9e electric SUVs gaining traction, Mahindra's share has climbed to around 17% in 2025, almost double what it was before. The sales data allows the brand to establish itself as the second leading electric vehicle manufacturer in the market. Mahindra also performs The three-wheeler EV market segment shows strong potential for the company because it currently leads this segment with a 41% market share. The company maintains a leading position in this particular market segment through its 41% market share. The company has expanded its electric vehicle product range.

MG Motor (JSW MG Motor) is another major player now. In multiple months of 2025, MG captured about 28–31% of the passenger EV market, which is quite impressive. It has become one of the biggest challengers to Tata, especially because MG's vehicles are priced competitively and have found a comfortable space in the urban EV market.

Two- and Three-Wheeler Segment **TVS Motor Company** continues leading the e-two-wheeler space with a stable share around 21–24% through most of 2025. TVS has benefited from a consistent product line and strong brand trust, which helps it stay ahead even as more companies enter the segment.

Bajaj Auto has seen a consistent growth trend with nearly 19% of electric two-wheelers' market share and 36

Ola Electric had more than 40% market share in 2024 but has now fallen to around 11–18% in September 2025 due to some production delays, supply chain problems, and regulations. In addition, traditional manufacturers have taken advantage of this problem to gain back their shares of the market.

Ather Energy also maintained its growth pattern, with a market share reaching up to 17% by September 2025. Ather Energy managed to keep its position owing to its high-quality engineering and product updates, as well as loyal customers.

Hero Electric (Hero MotoCorp VIDA) managed to gain some additional market share in 2025, reaching its share from 10% to 13% in the electric two-wheeler market. Hero's sub-brand of VIDA reached a 10.2% market share in July.

Foreign Companies in India's EV Market

MG Motor (JSW MG Motor) has grown so much that it now holds 28–31% of the passenger EV market. Its strategies, pricing, and product positioning have clearly resonated with Indian buyers.

BYD (China) has carved out a space mainly in the premium segment. It maintains a modest but steady market share of about 3–4% as of June 2025. BYD is more popular in the fleet and high-end markets.

Hyundai sits at around 3.9% market share as of mid 2025. It has been expanding its EV portfolio slowly, but not at the same pace as MG or Mahindra.

Mercedes-Benz, BMW, Volvo, Kia, Toyota, Citöen, Tesla all operate in niche, upper-tier segments. Together, they make up less than 5% of the EV market. Their buyers are mostly located in metro cities and usually fall under premium customer groups.

Key Market Trends

The electric car segment has witnessed many transformations within a short span. Tata, which was once dominating in the market, now holds reduced shares owing to the emergence of other brands such as MG and Mahindra. While international luxury brand cars have been making steady progress, their combined market shares are yet to make substantial contributions.

Electric two-wheelers constitute a fiercely competitive market. The established names in the industry like TVS, Bajaj, and Hero are competing against other emerging firms like Ather and Ola. It is often observed that the market leadership in electric two-wheelers changes frequently depending on various factors.

Mahindra and Bajaj dominate the market for electric three-wheelers. Some other brands, including Piaggio and Omega Seiki Mobility, have managed to create a niche for themselves but function in limited numbers.

Table 1.1: Market Share Summary Across All EV Segments (September 2025)

Segment	**Top Indian Companies**	**Market Share**
Cars/SUVs	Tata, Mahindra, JSW MG Motor	Tata – 34.14%, Mahindra – 15.76%, JSW MG Motor – 12.42%
Two-wheelers	TVS, Bajaj, Hero Electric, Ola, Ather	TVS – 21.24%, Bajaj – 15.18%, Hero – 11.28%, Ola – 11.17%, Ather – 9.37%
Three-wheelers	Mahindra, Bajaj	Mahindra – 14.56%, Bajaj – 13.14%, Piaggio – 13.16%, Omega – 11.87%

Current Competitive Structure (2025, Four-Wheeler EVs)

Table 1.2: Four-Wheeler EV Market Share in India (2025)

Brand	**Market Share (%)**	**Segment Focus**	**Trend**
Tata Motors	36–41	Mass market	Losing share, still leader
MG Motor India	28–31	Standard/SUVs	Surging, aggressive
Mahindra	15–22	Affordable SUVs	Gaining share
Hyundai	3.9–6	CUVs	Expanding slowly
BYD	3.4	Electric SUVs	Growing niche
BMW, Mercedes, Tesla, etc.	1–3	Premium/Luxury	Slow penetration

1.2.3 Electric Vehicle Battery Specifications in India (2025)

The number of EVs that exist in India has increased substantially since a few years back. In 2025, India has 55 EVs that range from scooters all the way up to heavy trucks. The following portion will discuss the battery parameters, battery chemistry, and voltages of the different EV models present. Out of the total EVs that are currently in use, 43 belong to the domestic market while 12 are imported. One aspect that should be highlighted is the electric bus model wherein BYD buses alone make up for roughly 2,330 buses in 35 different cities – which constitutes nearly one-fourth of the entire bus fleet.

Indian Companies EV Battery Specifications

Four-Wheeler Passenger Vehicles **Tata Motors** offers five EV models right now, each with its own battery configuration. The Nexon EV lineup consist of 30.2 kWh (320 V), 40.5 kWh (332.8 V) and 45 kWh (350 V) battery capacities which operate on LFP chemistry.. Meanwhile, the Tiago EV uses 19.2 kWh and 24 kWh battery packs at about 320 V, also based on LFP. Depending on the exact variant, these vehicles provide something between 312 and 489 km of driving range.

Mahindra has three EV models in this segment. The XEV 9e offers 59 kWh (350 V) and 79 kWh (400 V) batteries powered by NMC cells, and the BE 6 uses practically the same chemistry and capacity structure. A nice advantage is the fast-charging capability, 20% to 80% takes about 20 minutes. The range sits between 542 km and 682 km depending on configuration.

MG Motor (JSW) sells the Windsor EV, which comes in 38 kWh and 52.9 kWh packs (both at 350 V) using lithium-ion chemistry. The smaller Comet EV uses a 17.3 kWh LFP battery running at 300 V.

Two-Wheeler E-Scooters **TVS Motor** has several iQube variants, ranging from 2.2 kWh to 5.3 kWh at 52 V nominal. Their range can extend from 94 km to 212 km based on the pack size and operational environment.

Ola Electric S1 Pro models are fitted with 3 kWh, 4 kWh, or 5.2 kWh battery packs at 51.8 V. These use NMC/NCA 21700 cylindrical cells; the 4 kWh pack specifically uses a 14s16p arrangement.

Ather Energy uses 2.9 kWh and 3.7 kWh packs at 51.1 V nominal. These are based on NMC 811 chemistry—known for its high nickel content.

Bajaj Auto Chetak and **Hero MotoCorp VIDA** scooters both operate around 48–52 V and use batteries in the 2.9–3.5 kWh and 3.4 kWh ranges respectively.

Electric Buses (Operational) **Tata Starbus Urban** is among the most commonly deployed e-bus platforms. The 9 meter model uses a 186 kWh LFP battery, while the 12-meter AC version carries a 399.4 kWh pack, and both run at 665.6 V nominal. These buses typically deliver around 150–200 km in city conditions.

The Starbus Ultra City Electric (12-meter AC) uses a 213 kWh lithium-ion battery system paired with a 328 HP motor producing 3,000 N·m of torque. It can be fast-charged in roughly 2–3 hours.

Olectra Greentech BYD buses include the K9 (12-meter) with a 313 kWh battery at around 600 V and the C9 (9-meter) with a 200 kWh pack. More than 2,330 units of these buses operate across India, making them a major part of city transit.

PMI Foton runs midi buses with 150 kWh LFP batteries and mini buses with 102 kWh packs, both generally around the 400 V class.

Switch Mobility has a standard electric bus equipped with a 389 kWh LFP pack at 400 V.

JBM Solaris features a 200 kWh NMC battery (400 V nominal) capable of delivering about 250 km of range per charge.

Electric Trucks and Commercial Vehicles (Operational) **Tata Motors** has four EV trucks currently operating:

- **Ace EV:** A light commercial truck with a 21.3 kWh LFP battery (400 V), delivering around 154 km range and powered by a 21 kW motor.

- **Ultra E.9:** Comes in 110 kWh and 148 kWh LFP variants (400 V). The truck uses a 335 HP motor with 950 N·m torque and supports fast charging within 1.5–2 hours.
- **Prima E.28K:** Uses a massive 453 kWh LFP battery at 400 V, suited for demanding use cases. It has a 245 kW motor producing 2,950 N·m torque.

Ashok Leyland offers two EV trucks:

- **Boss 1219 EV:** 80 kWh battery at 400 V, offering a 150 km range with a 187 HP motor.
- **Boss 14 HB EV:** Uses a larger 201.5 kWh pack at 400 V, with a 160 HP motor capable of going about 230 km.

Switch Mobility also provides:

- **IeV3:** Comes with a 25.6 kWh (400 V) battery capable of 90 km.
- **IeV4:** Has a 32.2 kWh pack (400 V), and a 60 kW motor giving 230 N·m torque and around 120–125 km of range.

Eicher Motors offers the **Pro 2055 Electric**, available in 64.4 kWh and 70 kWh packs, both at 400 V, delivering 120–162 km.

Omega Seiki has the **OSM M1KA** fitted with a 90 kWh 400 V battery and a 30 kW motor giving 347 N·m torque. Its range is about 80 km with a 1-tonne payload.

Foreign Company EV Battery Specifications (Operational in India)

Passenger Four-Wheelers **BYD Atto 3** uses BYD's Blade LFP battery in 49.92 kWh and 60.48 kWh versions at 400 V. It supports DC fast charging of 70–80

Table 1.3: Battery Ratings and Nominal Voltages of Most Popular Indian Manufacturer Electric Vehicles

Manufacturer	Model Name	Battery Capacity (kWh)	Nominal Voltage (V)	Vehicle Type
Tata Motors	Nexon EV	40.5	332.8	Sedan Utility Vehicle
Tata Motors	Nexon EV	45	350	Sedan Utility Vehicle
Tata Motors	Tiago EV	24	320	Hatchback
Mahindra	XEV 9e	79	400	Sedan Utility Vehicle
Mahindra	BE 6	59	350	Sedan Utility Vehicle
MG Motor (JSW)	Windsor EV	52.9	350	Crossover Utility Vehicle
TVS Motor	iQube	3.5	52	Electric Scooter
TVS Motor	iQube Standard	5.3	52	Electric Scooter
Ola Electric	S1 Pro	4	51.8	Electric Scooter
Ather Energy	Ather 450X	3.7	51.1	Electric Scooter
Bajaj Auto	Chetak	3.5	48	Electric Scooter
Hero MotoCorp	Vida V1	3.4	52	Electric Scooter
Tata Motors	Starbus Urban	399.4	665.6	City Bus Alternating Current (12-meter)
Olectra Greentech (BYD)	BYD K9	313	600	City Bus (12-meter)
Tata Motors	Ace EV	21.3	400	Light Commercial
Tata Motors	Ultra E.9	148	400	Medium Commercial
Tata Motors	Prima E.28K	453	400	Heavy Commercial
Ashok Leyland	Boss 1219 EV	80	400	Medium-Duty

kW, allowing 0–80% charging in roughly 45 minutes. The range is around 468–521 km.

Hyundai Kona Electric runs on a 39.2 kWh NMC-based pouch battery (approx. 350 V) that delivers up to 452 km.

BMW iX1 is equipped with a 66.4 kWh prismatic lithium-ion pack (around 400 V), offering about 531 km.

Mercedes-Benz EQS features a large 107.8 kWh battery (400 V), using pouch cells and offering up to 857 km range.

Kia EV6 uses an 800 V architecture coupled with an 84 kWh battery (77.4 kWh usable). It supports extremely fast charging, 10–80% in about 18 minutes using a 350 kW charger.

Electric Buses (Operational in India) **BYD buses** operating under Olectra Greentech include:

- **K9 (12-meter)** with a 313 kWh LFP battery at ∼600 V and around 250 km range.
- **K9 (7-meter)** using 215–254 kWh packs at similar voltage.

- **C9 Tourist (9-meter)** carrying a 324 kWh LFP pack designed for intercity use.

By January 31, 2025, about 2,330 BYD-equipped buses were running in 35 cities, collectively covering over 335.2 million km. One of these buses alone logged 678,109 km, showcasing battery durability.

Table 1.4: Battery Ratings and Nominal Voltages of Most Popular Foreign Manufacturer Electric Vehicles Operating in India

Manufacturer	**Model Name**	**Battery Capacity (kWh)**	**Nominal Voltage (V)**	**Vehicle Type**
BYD (China)	Atto 3	60.48	400	Sedan Utility Vehicle
Hyundai (Korea)	Kona Electric	39.2	350	Sedan Utility Vehicle
BMW (Germany)	iX1	66.4	400	Sedan Utility Vehicle
Mercedes-Benz (Germany)	E-Class Sedan	107.8	400	Sedan
Kia (Korea)	E-Vehicle 6	84	697	Sedan Utility Vehicle
BYD (China)	K9 (12-meter)	313	600	City Bus (12-meter)
BYD (China)	C9 Tourist	324	600	Tourist Bus (9-meter)

This research was conducted by reviewing [20–30].

1.3 Motivation and Problem Formulation

The rising popularity of electric cars and their demand for rapid charging facilities has created a necessity for better charging methods. The first thing that needs to be addressed while planning charging stations is the safe exchange of energy between the power source and the battery of the EV. This is because the EV itself has connections directly to the battery, and any leakage during the process will be dangerous. Therefore, isolation between input and output stages becomes a crucial issue for converters.

Unlike most DC-DC converters such as buck and boost converters, electrical isolation between the input and output does not exist in most conventional DC-DC converters. Although such converters are highly efficient and effective in their purpose, they cannot be used in critical applications such as power transfer in EVs. Additionally, the lack of isolation in such converters causes issues such as failure propagation, difficulties with safety, and challenges in satisfying the necessary regulations. Therefore, isolated DC-DC converters must be employed to solve this problem. Isolation in such converters is usually provided through the use of transformers. Besides ensuring the safety of the system, the use of transformers also provides the opportunity for flexible voltage scaling depending on the turns ratio of the transformer. Hence, isolated converters have an advantage in applications such as EV charging since there are different voltages on both sides of the converter.

Despite the variety of isolated converters, one of the most interesting converters today is the Dual Active Bridge (DAB). Unlike other isolated converters, the DAB converter has the ability to operate as a bidirectional converter with high efficiency. The converter operates with higher frequencies than other isolated converters, resulting in reduced dimensions of the passive components used in the circuit. Also, ZVS switching makes DAB converters highly efficient. Nevertheless, designing DAB converters involves many complications and requires careful consideration of transformer parameter choice, leakage inductance, and proper switching conditions. Designers face challenges regarding the maintenance of high efficiency levels over a wide range of operating conditions. Hence, there is a need to conduct studies related to the design of isolated converters and investigate the characteristics of such converters.

For the purposes of this project, a dual-active bridge converter whose gate driver circuit is controlled by an FPGA board will be designed. The objective of the research is to design a hardware model of the DAB converter based on FPGA-controlled gating signals.

Chapter 2

DAB Converter and Two Stage Converter

2.1 Overview of DAB converter

A DAB converter, or a dual active bridge converter, is a power electronic converter that consists of two DC to AC converters coupled using a high-frequency transformer. It actually works as a DC to DC converter where, using the turns ratio, we can decrease or increase the output voltage for the load side appropriately according to the application. The IGBTs or MOSFETs used in the converter are switched such that one pair of switches operates for 50% of the duty cycle and the other pair operates for the remaining 50% of the duty cycle on the primary side. Similarly, on the secondary side, the same switching scheme is applied, but the primary side switches are phase-shifted by a certain angle ahead of the secondary side switches. An inductor is also present, and its purpose is to enable power transfer from the primary side to the secondary side.

The output power of a DAB converter is given by:

$$P_o = \frac{V_{dc1} V_{dc2} N \phi (\pi - |\phi|)}{2 f_{sw} L \pi^2} \tag{2.1}$$

where V_{dc1} is the primary side voltage, V_{dc2} is the secondary side voltage, N

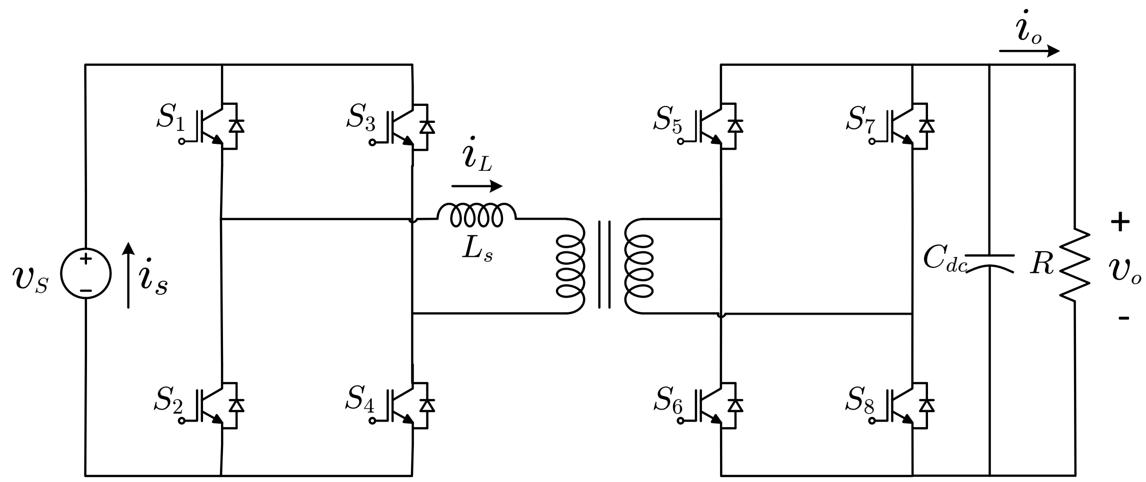


Figure 2.1: DAB Converter Schematic

is the turns ratio, ϕ is the phase difference between switching pulses of the primary and secondary, f_{sw} is the switching frequency. The schematic of the converter is shown in Fig. 2.1.

The DAB converter is widely used in applications where bidirectional power flow is required, such as electric vehicle charging, battery energy storage systems, and DC microgrids. One of the main advantages of this topology is that it provides galvanic isolation through the high-frequency transformer while still maintaining high efficiency and power density [10].

The operation of the DAB converter is based on the phase shift between the primary and secondary bridge voltages. When there is no phase shift between the two bridges, ideally no net power is transferred. As the phase shift increases, the power transferred also increases, reaching maximum at a phase shift of $90°$. The direction of power flow depends on which bridge leads the other. The current flowing through the inductor is typically trapezoidal in nature due to the square wave voltages generated by the bridges. This results in a high-frequency AC current in the transformer, which is then rectified back to DC on the secondary side. The shape and magnitude of this current depend on the phase shift, input/output voltages, and inductance value [11].

Another important feature of the DAB converter is its ability to achieve soft switching, particularly Zero Voltage Switching (ZVS), over a wide operating range. Thus, a significant reduction in switching losses is achieved and the system efficiency increases, especially at high frequencies of switching processes. In addition, since ZVS operation is provided, the stresses on semiconductor devices decrease; therefore, the reliability of the converter is increased [13].

The leakage inductance of the transformer has a great importance in the operation of the DAB converter. As distinct from conventional transformers, where leakage inductance should be minimized, in the case of the DAB converter, the latter is actively used as an energy carrier. Thus, the magnitude of leakage inductance determines the amount of transferred power, current waveform, and conditions of achieving the soft switching mode. One more advantage of the DAB converter is associated with its ability to operate at high frequencies. Being operated at high frequencies, a transformer becomes quite smaller than low-frequency transformers, thus allowing obtaining high power density of the device [14].

At the same time, there are also disadvantages related to the DAB converter. For example, the appearance of circulating and reactive power can lead to increased conduction losses depending on the control strategy and operational modes [31].

Table 2.1: Simulation parameters used for the DAB converter

Parameter	Value
Primary Side Voltage, V_{dc1}	48 V
Secondary Side Voltage, V_{dc2}	120 V
Switching Frequency, f_{sw}	25 kHz
Transformer Turns Ratio, N	20/48
Leakage Inductance, L_s	55 μH
Load Resistance, R	100 Ω
Output Capacitance, C_{dc}	470 μF

2.2 Open Loop Simulation of DAB Converter

2.2.1 Simulation Parameters

2.2.2 Phase Shift Calculation and Power Transfer Characteristics

To analyze the ability of the power transfer capability of the DAB converter, the relation between the output power and phase shift was utilized. In particular, for a specified pair of input and output voltages, switching frequency and inductance, the output power is a function of the phase shift angle that exists between the two bridges of switching signals.

For that reason, the output power is first determined through the use of the well-known formula that gives the power transfer capability of the DAB converter at various values of phase shifts that exist between $0°$ and π. This way, the power transfer characteristic of the converter can be identified. Moreover, the reference value of output power was defined based on the resistive load and the output voltage. With the use of this reference value, the phase shift angle that corresponds to the reference value can be determined by arranging the power equation. As a result, there are two values that satisfy the equation, where the smaller phase angle is chosen due to lower current stress. Eventually, the power vs phase shift graph together with the reference output power and the phase shift that corresponds to it were depicted. The graph of power transfer characteristics of the DAB converter concerning the phase shift angle is depicted in Fig. 2.2. For the parameters presented in Table 2.1, the phase shift value is $37.52°$.

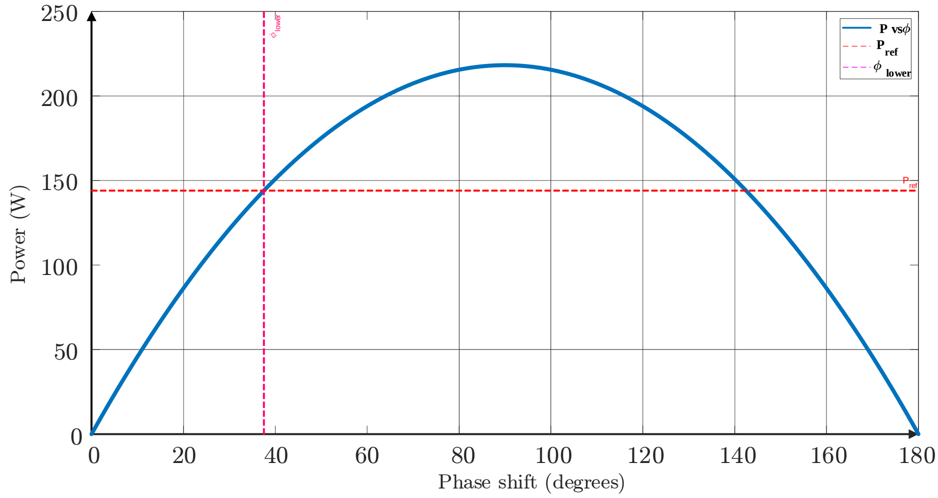


Figure 2.2: Power Transfer Curve for DAB Open Loop Smulation

2.2.3 Simulation Setup

An open-loop simulation of the DAB converter has been performed in this work using MATLAB/Simulink. The circuit used two full bridges linked by a high-frequency transformer. The bridges were made up of IGBTs. The simulation results have been obtained using the values shown in Table 2.1. In this simulation, each full bridge has been operated in a switching pattern with a 50% duty cycle. Diagonal switches were fired such that they operated together while implementing the switching pattern. Also, the switches operating at the primary side and secondary side have been operated with a phase shift such that power is controlled by how much this phase shift value is changed. The transformer in this simulation has been considered as an ideal transformer with the given turn ratio, and an external inductance of 55 μH is taken into account to consider the effect of leakage inductance. As per the design of DAB, the inductance used here acts as the most important energy storage element and plays a key role in determining the current waveform and transferring power. However, in practical applications, some part of this inductance may be caused by transformer leakage. In the output, a resistor and a capacitor have been taken into account for load and smoothing purposes, respectively.

This was a simple open-loop simulation where the required phase shift for transferring power from the input stage to the output stage was fixed depending on the required output power. No control mechanism was used for controlling any of the variables. Different signals have been observed including switching pulses, inductor current, input current, and output voltage, etc.

2.2.4 Simulation Results

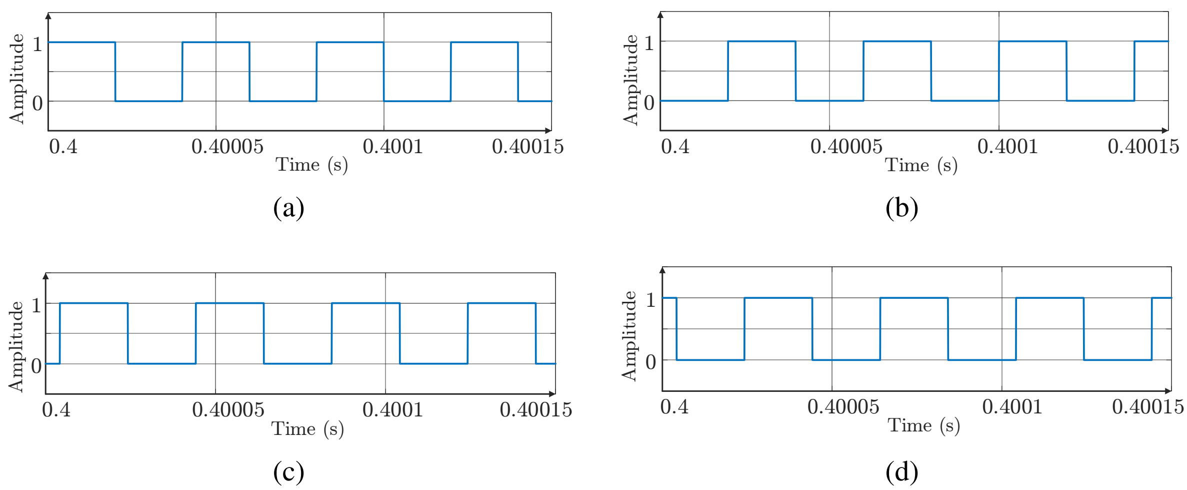


Figure 2.3: (a) Switching Signal for S_1 and S_4 (b) Switching Signal for S_2 and S_3 (c) Switching Signal for S_5 and S_8 (d) Switching Signal for S_6 and S_7

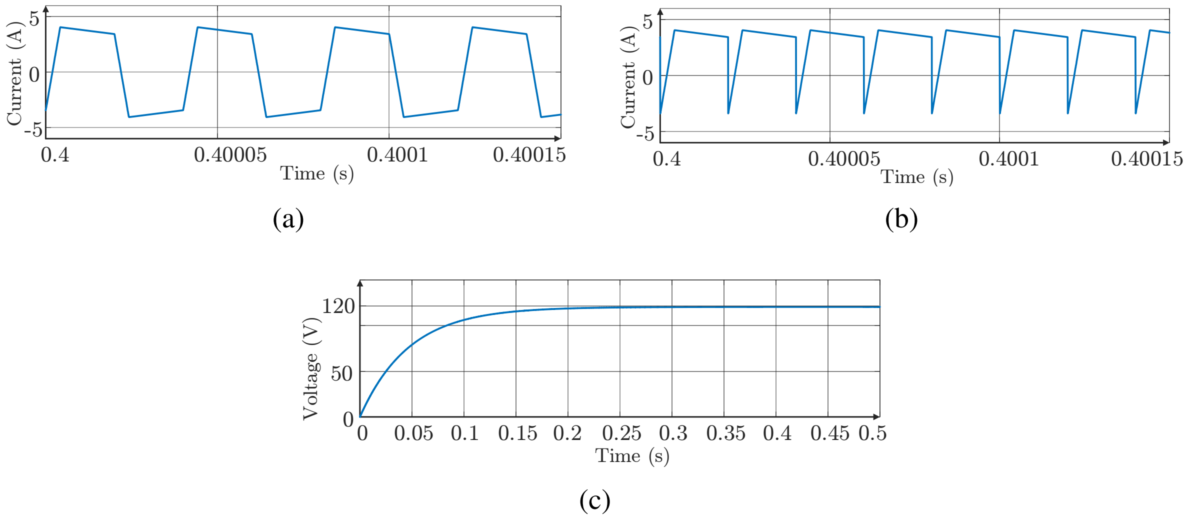


Figure 2.4: (a) Inductor Current (b) Input Current) (c) Output Voltage

2.2.5 Observation and Discussion

The inductor current from simulation results gave a maximum current of about 4.04 A and a nearly zero average current. It is expected to observe such currents because in the DAB converter, there is an alternating current in the inductor, facilitating energy flow from the primary side to the secondary side. The obtained graph (Fig. 2.4(a)) demonstrates symmetric behavior around zero, thus indicating successful operation of the converter. The average input current was recorded as 3.038 A while the output voltage had a value of approximately 120.4 V. The output voltage was found to be nearly constant owing to the presence of the output capacitor. The output power to the load was 144.96 W, while the input power was measured at around 145.82 W. These indicate good efficiency in the operation of the converter. The converter efficiency can be calculated as:

$$\eta = \frac{P_{out}}{P_{in}} \approx \frac{144.96}{145.82} \approx 99.4\%$$

The main factor behind this efficiency is the use of high frequencies and soft-switching techniques in the DAB converter. The second major point observed from the input current waveform is that the current assumes negative values at certain intervals. This shows that there is bi-directional flow of power in the circuit, which is one of the main benefits of using the DAB converter. It can be used in various applications including battery charging and discharging in electric vehicles.

From the results above, it is clear that the simulation was successful and that the performance of the DAB converter matches its theory.

2.3 Closed Loop Control of DAB Converter

2.3.1 Introduction and Control Objective

In Section 2.2, the analysis of the Dual Active Bridge (DAB) converter was carried out in the open-loop scenario where a fixed phase shift had to be used to transfer power from source to the load. Although such an open-loop approach may help understand the fundamental working of the converter, the output of the circuit cannot be guaranteed to remain regulated when subjected to load or input variations. As such, closed-loop control is essential to regulate the output voltage of the system.

The output power of a DAB converter can be varied by changing the amount of phase shift between the primary and secondary bridge signals. In that regard, a closed-loop scheme would entail the regulation of the phase shift as a function of the voltage error obtained between the reference and measured voltages. Thus, the goal of this project is to regulate the output voltage of the DAB converter to a set reference voltage level through the regulation of phase shift. The difference between the measured output voltage and the reference voltage will be compared and then processed by a controller to determine the appropriate phase shift. To facilitate the design process of the controller, a small signal model of the DAB converter is derived. The relationship between the output voltage and phase shift in the DAB converter is given by

$$G_{v\phi}(s) = \frac{\hat{v}_2}{\hat{\phi}} = \frac{NV_1(1-4\Phi)}{f_s L} \frac{R_L}{R_L C_2 s + 1} \tag{2.2}$$

where N is the turns ratio of the transformer, V_1 is the voltage applied to the circuit, f_s stands for the switching frequency, L is the series inductance, R_L is the load resistance, and C_2 is the capacitance of the output.

The phase shift ϕ is limited to the interval $-0.25 \leq \phi \leq 0.25$. The above transfer function allows a basis to design controllers. Several similar approaches based on small-signal models have been reported in the literature for the purpose of designing controllers [11, 12].

Hence, using closed-loop control, DAB converters can be made capable of producing regulated voltages, dynamic responses, and robustness.

2.3.2 Control Strategy and Controller Design

For the present case of DAB converter's closed loop operation, Single Phase Shift (SPS) control technique is used. In the SPS control, the primary and secondary bridges switch with 50% duty cycle, while the power is transferred by applying a phase shift between them. The phase shift is applied to the switching signal of the secondary bridge, whereas the primary switches operate in the same 50% duty cycle state. SPS control is easy to apply and is one of the most popular control strategies used in practice [10, 13].

The voltage feedback loop was designed to stabilize the output voltage. The difference between the output voltage measured during the simulation and

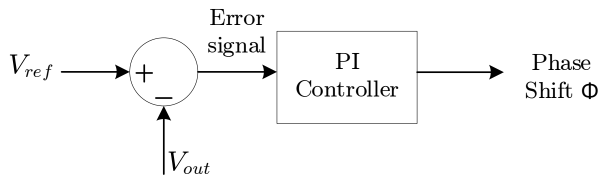


Figure 2.5: Voltage Control Loop for DAB Converter

the voltage reference was processed with a Proportional-Integral (PI) controller. The controller's output gave an indication about the necessary phase shift needed to be added to the switching signal of the secondary bridge (see Fig. 2.5). For the design of the PI controller, the small-signal transfer function obtained in the preceding section was used. With the help of frequency domain analysis, an initial set of gains was computed. The selection criteria for the gains K_p and K_i was based on choosing the magnitude of the system's open-loop transfer function equal to unity at the chosen crossover frequency. Thus, in the MATLAB software, the gains were obtained by calculating the system transfer function at a specified crossover frequency. Those gains served as a good approximation and could be further tuned to get better results. Tuning of the gains was done using simulations, which led to the improvement in the dynamic behavior of the controlled system. After computing the initial values of the gains, the PI controller was adjusted to minimize the settling time and ensure good performance. As a result of tuning, the response of the system became much faster and stable, with the settling time being significantly reduced. To examine the robustness of the controller, a sudden change in the load resistance from 100 Ω to 150 Ω was simulated.

Thus, it may be concluded that the developed combination of SPS control strategy and feedback system provides efficient voltage regulation with simple implementation.

2.3.3 Simulation Setup

The closed loop simulations for the proposed DAB converter design were done using MATLAB/Simulink. For the purpose of this simulation, most of the values presented in Table 2.1 were used but the resistance of the load was adjusted to be different from the one given in the table. The load was adjusted from 100 Ω to 150 Ω to investigate the dynamic performance of the system. In the design of the converter system, a closed-loop voltage controller design was used. That is, the output voltage was sensed and compared with the reference voltage. Then the difference between the reference voltage and the measured voltage (the error signal) was supplied to a PI controller. The output of the controller is used to produce the necessary phase angle, which is used to adjust the phase shift of the secondary side switching pulses. The proportional-integral gain values were initially estimated using MATLAB analytically. This was done by writing a MATLAB program for computing the appropriate values of K_p and K_i using the transfer function of the system and the crossover frequency. The tuning process for the gains was further done using trial and error. The finally obtained gains of the controller were $K_p = 0.095$ and $K_i = 16$.

The switches were modeled as IGBTs for both primary and secondary bridges. Ideal transformers were used for isolation together with a series inductance for transferring the power between the bridges. A resistive load with an output side capacitor for smoothing was used on the secondary side. The primary switching gating pulses had constant duty cycles while the secondary pulse trains had phase shift depending on the PI controller output.

2.3.4 Simulation Results

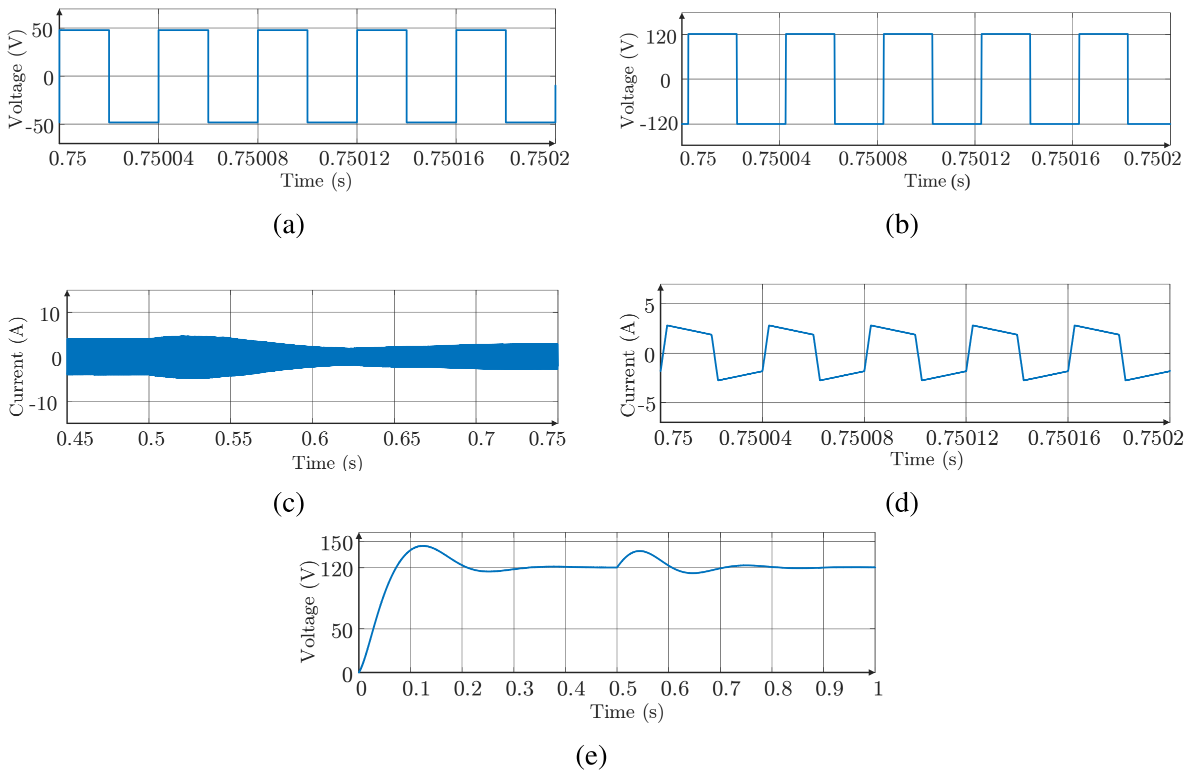


Figure 2.6: (a) Primary Bridge Voltage (b) Secondary Bridge Voltage (c) Inductor Current Transient (d) Inductor Current in Steady State (e) Load Voltage

2.3.5 Observation and Discussion

According to the simulation results obtained from the closed-loop simulation, the converter was running in steady-state operation with an average input current of 2.99 A prior to the load variation. The maximum inductor current was about 3.98 A while the output voltage was approximately equal to 119.9 V, close to the desired reference value. The steady-state output voltage waveform (Fig. 2.6(e)) is shown to be well-regulated, thus demonstrating the correct behavior of the control system. After varying the load resistor from 100 Ω to 150 Ω, the system showed its transient behavior. During this time period, the inductor current changed as expected and was properly regulated according to the requirement, shown in Fig. 2.6(c). As soon as the system settled to the new steady state, the input current became approximately 2 A and the peak inductor current reached the value of 2.59 A. The output voltage, on the other hand, was stabilized at approximately 120 V. According to the steady-state inductor current waveform (Fig. 2.6(d)), the system achieved steady-state operation after varying the load resistance. Moreover, the inductor current in steady-state condition exhibits some kind of DC offset. However, the magnitude of the offset is in the milliampere range, which makes it negligible when compared to the total system current. High efficiency can be said to characterize the converter operation under load variation. Prior to changing the load resistor, the efficiency was 99.8%, whereas, after varying the load resistor, it was close to 100%. The efficiency of the converter in this case has been found high because of the operation frequency and the soft-switching operation. An important point about the converter operation is that there exist negative peaks in the input current waveform. Thus, it can be concluded that the converter supports bidirectional power transfer, one of the advantages of the DAB converter.

Overall, the simulation result has proved to be promising regarding the voltage regulation capability of the converter under load variations.

2.4 Overview of Two-Stage Converter

Two-stage converter architectures are frequently used in contemporary power electronics applications to provide improved power quality, voltage regulation, and isolation. Typically, the two-stage converters include both AC-DC converters and DC-DC converters, where each stage serves a certain purpose. In accordance with its application needs, various types of two-stage converters can be developed. In the current work, the configuration of a two-stage converter that includes an AC-DC front-end together with a

DC-DC converter was considered. The general structure of the proposed two-stage converter is presented in Fig. 2.7. The first stage is represented by an AC-DC rectifier with a boost converter. The PFC-based operation of the boost converter is ensured by utilizing unit vector template control. Thus, the first stage is responsible for the transformation of AC input supply into a regulated DC output voltage along with shaping the input current into the sinusoidal shape. Typically, the AC-DC rectifiers draw non-sinusoidal currents from the input source, thus creating problems with low power factor and presence of harmonics. Such harmonics are detrimental for the whole system, causing extra losses, overheating, and decreasing the efficiency of the system. Therefore, some techniques of improving power factor and reducing harmonic content are often implemented in the circuits. In the current work, the unit vector template control in the boost converter helped improve the power factor and eliminate harmonics.

The second stage of the converter includes a Dual Active Bridge (DAB) converter. This type of converters was described earlier. The DAB converter provides galvanic isolation between the converter and the load as well as controls the output voltage level. Moreover, the DAB converter allows the system to transfer power in both directions, which increases the applicability of such systems for battery charging and discharging.

The entire two-stage converter system composed of the rectifier, PFC boost converter, and DAB converter is presented in Fig. 2.7. As it can be observed from the figure, the front-end stage is capable of providing high-quality power at the input.

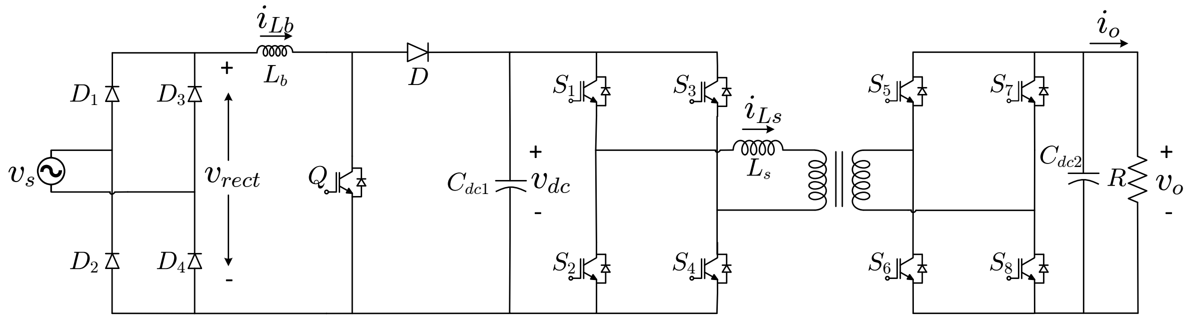


Figure 2.7: Two Stage Converter Schematic

2.5 Two Stage Converter with Open Loop DAB

2.5.1 System Configuration and Operation

The two-stage converter used in this project includes an AC-DC front end and a Dual Active Bridge (DAB) converter operating in the open-loop. The overall structure ensures power factor correction in the input stage and isolated DC-DC conversion in the output stage.

At the input stage, AC power is rectified and boosted using a boost converter with the PFC scheme implemented via the unit vector template control method. This ensures sinusoidal input current and its phase alignment with the source voltage. As a result, the power factor is improved while harmonics are suppressed. On the other hand, the boost converter controls the DC link voltage at the required level for the second stage. The latter utilizes the DAB converter that has already been described in Section 2.1. In the present case, the DAB converter operates in the open-loop, where the fixed phase shift is provided between the primary and secondary bridge switching signals. The phase shift determines the power transferred from the former to the latter. Due to the open-loop operation, there was no feedback control applied to the output voltage; therefore, it depends only on the phase shift value and system parameters. Power is delivered from the AC source to the rectifier and PFC boost converter, further providing the DC link voltage. Then, it is transferred to the load via the DAB converter. Utilization of a high-frequency transformer in the DAB stage provides galvanic isolation and effective power delivery.

Thus, the overall two-stage scheme guarantees power quality improvement at the input stage and controlled power delivery in the output stage.

2.5.2 Control Strategy of Front-End Converter

The Unit Vector Template (UVT) control technique was applied to the front-end converter as illustrated in Fig. 2.7. The proposed technique allows shaping the input current according to the shape of the input voltage and thus improves the power factor and mitigates harmonic contents in the input current.

For this purpose, the output voltage of the diode bridge rectifier was measured and used as a template for creating a unit vector of reference current. This was done by normalizing the measured rectifier voltage with its peak value and thus obtaining a waveform representing the input voltage shape. Next, the DC-link voltage of the rectifier was measured and compared with the reference value. The difference between them is processed by a proportional-integral controller in order to get the peak value of the reference current. Then, the peak value obtained from the controller is multiplied with the template unit vector to obtain the reference instantaneous current waveform. The measured reference current is next compared with the inductor current of the boost converter. The difference between both currents is passed through a hysteresis current controller in order to compare them. The hysteresis current controller is characterized by a predefined bandwidth

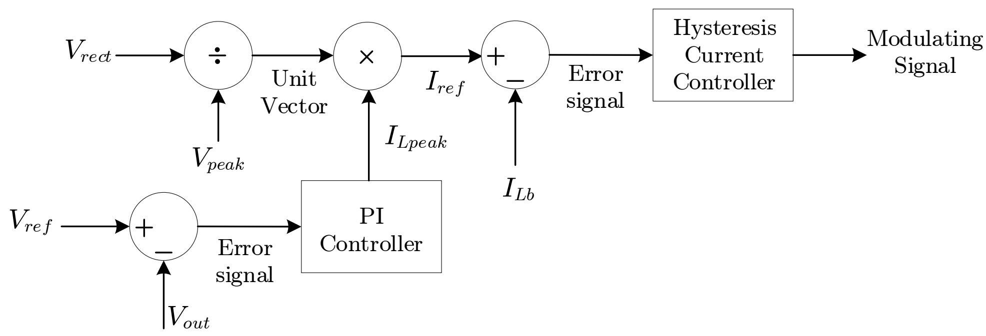


Figure 2.8: Block Diagram for Unit Vector Template Control Strategy

within which it ensures the proper tracking of the actual current with the reference current. In this way, based on the error obtained between the two currents, switching pulses are obtained for the boost converter switch Q. As mentioned above, this technique helps to control the input current in such a way that it becomes sinusoidal and aligned with the input voltage, which will provide unity power factor of operation of the converter. The employment of hysteresis current control provides a fast dynamic response, making it possible for the converter to respond quickly to any changes in the loading conditions and input characteristics. Due to the fact that there are no any transformations or synchronization techniques required in the process of controlling the input current, the proposed technique is easy to implement. The UVT control strategy has been widely analyzed and discussed in literature [32–34].

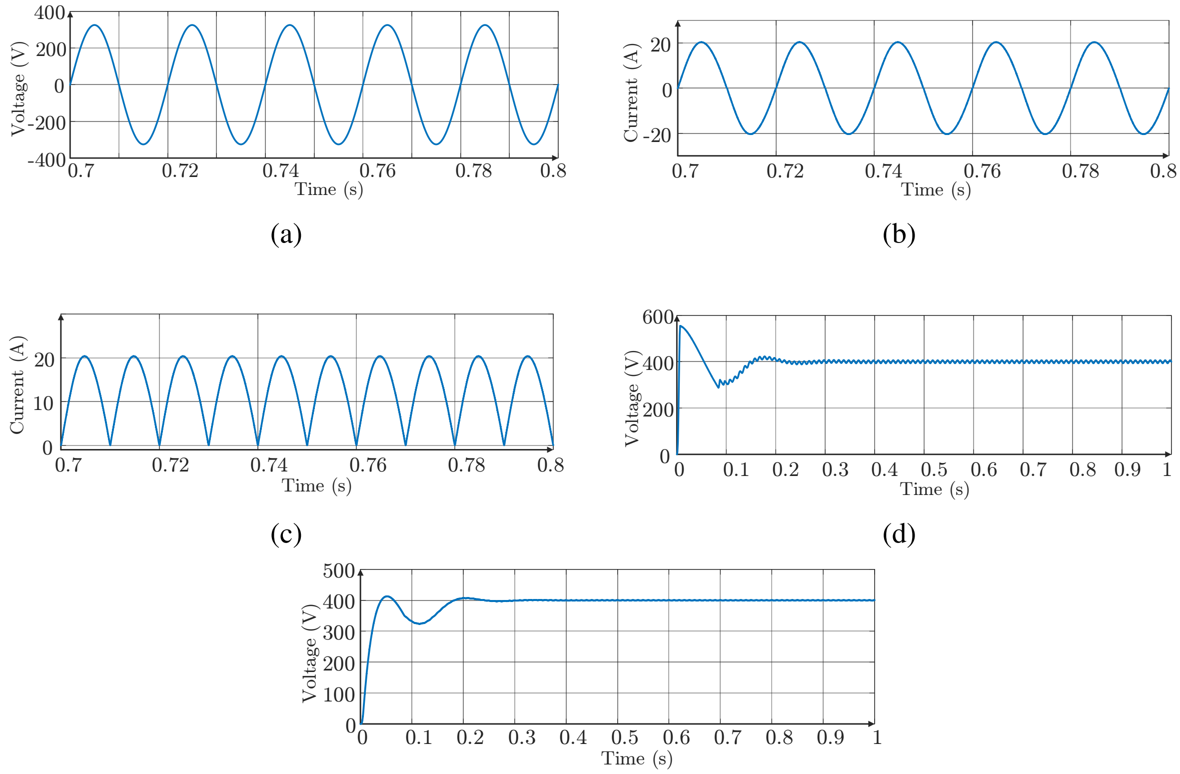
Fig. 2.8 presents the structure of the control algorithm.

2.5.3 Simulation Parameters

Table 2.2: Simulation parameters for the two-stage converter system

Parameter	**Value**
Supply Voltage, V_s	230 V (RMS)
Supply Frequency, f	50 Hz
Boost Inductance, L_b	1.2 mH
DC Link Capacitance, C_{dc}	2 mF
DC Link Voltage, V_{dc}	400 V
DAB Switching Frequency, f_{sw}	10 kHz
Transformer Turns Ratio, N	1:1
DAB Series Inductance, L_s	0.5 mH
Load Power, P_{rated}	3.2 kW
Load Resistance, R	50 Ω
Output Voltage, V_o	400 V

2.5.4 Simulation Results



(a) (b) (c) (d) (e)

Figure 2.9: (a) Source Voltage (b) Source Current (c) Current Through L_b (d) DC Link Voltage (e) Load Voltage

2.5.5 Observation and Discussion

The simulation results revealed that the RMS value of the source voltage was around 230 V while the source current RMS value was approximately 14.29 A. The waveform of the input current was similar to that of the supply voltage waveform, hence showing effective operation of the front-end converter.

Figures 2.9(a) and 2.9(b) illustrate that the source current lags behind the source voltage phase by zero, implying that there is unity power factor. It further shows the effectiveness of the unit vector template controller in controlling the input current, thus minimizing harmonics. The steady state DC link voltage was around 400 V while the load voltage was approximately

399.9 V.

Zero-phase difference between source current and voltage indicates the unit vector template controller has made possible operation at unity power factor. The front-end power factor corrector ensures that the power quality of the supply line is improved by ensuring sinusoidal current draw from the input power source. Under these conditions, the converter achieves an efficiency level of 97.3%. This implies the efficiency of the system is relatively high due to minimized switching losses and efficient transfer of power via the DAB converter.

2.6 Two-Stage Converter with PFC and Closed-Loop DAB Control

2.6.1 System Overview

The performance of the two-stage converter with power factor correction and closed-loop DAB control is investigated in this part. The configuration was more or less similar to the one before, having the AC-DC front-end coupled with the DAB converter.

For the front-end stage, a UVT-based power factor correction was employed for the boost converter. As a result, the sinusoidal input current was achieved with the use of power factor correction technique, making sure that the input current was in phase with the voltage across the input terminals. In the second stage, voltage-based closed-loop control strategy was implemented, as mentioned above in Section 2.3.2. Output voltage control relied on the PI controller, providing the necessary phase shift between primary and secondary bridges.

To summarize, the combination of both input power factor correction and output voltage regulation was considered here.

2.6.2 Simulation Setup

For two-stage charging including the closed-loop control of the DAB converter, the simulation of the system in MATLAB/Simulink has been performed. Almost all of the parameters are similar to those in Table 2.2 with some exceptions.

The capacitance at the output was taken as 2 mF. To analyze the impact of changing conditions on the control process, the load resistance was varied from 100 Ω to 200 Ω during the simulation. In this case, the DAB series inductance was set as 55 μF compared to its value in Table 2.2. The power factor correction was provided by means of the PI controller ($K_p = 2$, $K_i = 10$). For the DAB converter, the PI voltage control is used ($K_p = 0.05$, $K_i = 1.58$).

The simulation ran long enough to show how the system responded during changes and after things settled.

2.6.3 Simulation Results

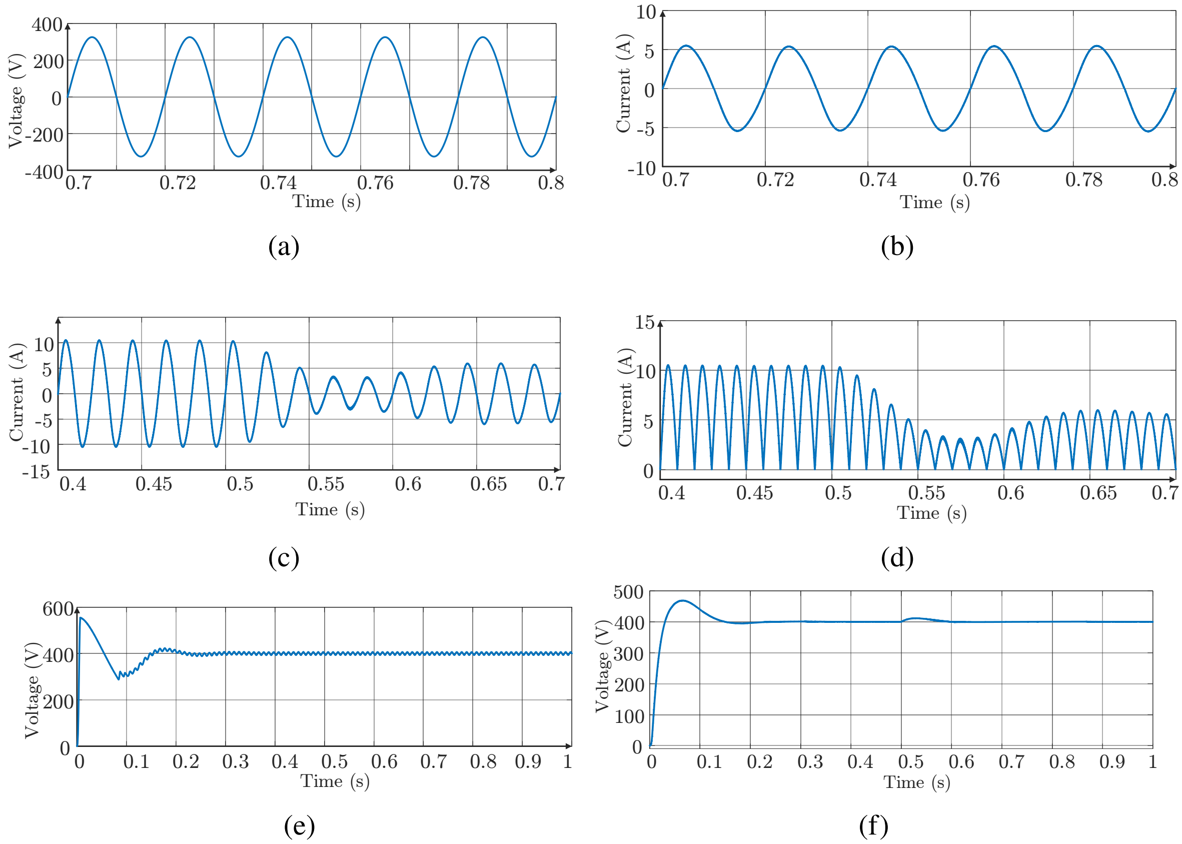


Figure 2.10: (a) Source Voltage (b) Source Current (c) Source Current Transients at Load Change (d) Current Through L_b at Load Change (e) DC Link Voltage (f) Load Voltage

2.6.4 Observation and Discussion

Results show that the RMS value of the source voltage is close to 230 V. In case the load resistance rises from 100 Ω to 200 Ω, the source RMS current drops from around 7.33 A to 3.8 A. Moreover, the output voltage is always kept near 400 V even during switching between different loads. The analysis of Fig. 2.10(a-b) shows the absence of any phase shift between the source voltage and current values, which means that their values are almost coincident. As a consequence, the power factor approaches unity.

Consequently, the measurement results prove that the proposed system is capable of providing stable voltage regulation at various loads. High voltage regulation capability is an indicator of proper functioning of the closed-loop

DAB control strategy. An increase in load resistance is followed by a proportional drop in the input current according to the circuit theory fundamentals. Furthermore, a small phase shift between the source voltage and current values demonstrates that the power factor correction stage works effectively. The high efficiency of the converter before and after the load variation confirms that the designed control method enables the efficient operation of the converter and high-quality input power.

In conclusion, the combination of UVT-based power factor correction and closed-loop DAB control improves the quality of input power while providing stable output voltage regardless of changes in the load.

Chapter 3

Hardware Implementation of DAB Converter

3.1 Introduction

Earlier chapters discussed the operation of the Dual Active Bridge (DAB) converter through simulations under both open-loop and closed-loop conditions. Even though simulations give a great deal of information about how the converter functions, the verification of the theoretical approach and experimental assessment are critical for the hardware implementation of the converter.

This chapter is dedicated to the hardware implementation of the converter. The primary goal is to build a model prototype and assess its performance at both open loop and closed loop operation. The hardware implementation will include designing the power circuitry, selecting appropriate components, creating a gate driver, and generating driving signals via an FPGA. Unlike simulations, where all the parameters can be theoretically modeled, actual implementation will involve switching losses, parasitic phenomena, PCB and magnetic components limitations, and components non-idealities.

The following sections describe in detail the hardware implementation of the converter. Experimental results obtained from testing the prototype are also discussed and analyzed.

3.2 Hardware Design and Component Selection

3.2.1 PCB Design and Implementation

The PCB of the DAB converter was designed using Altium Designer. The PCB layout has been designed keeping in mind the high current paths, spacing between tracks and adequate copper area so that the circuit is capable of handling the desired current without overheating. Adequate care has also been taken to keep the PCB design simple and compact. The entire PCB assembly of the H-bridge is presented in Fig. 3.1.

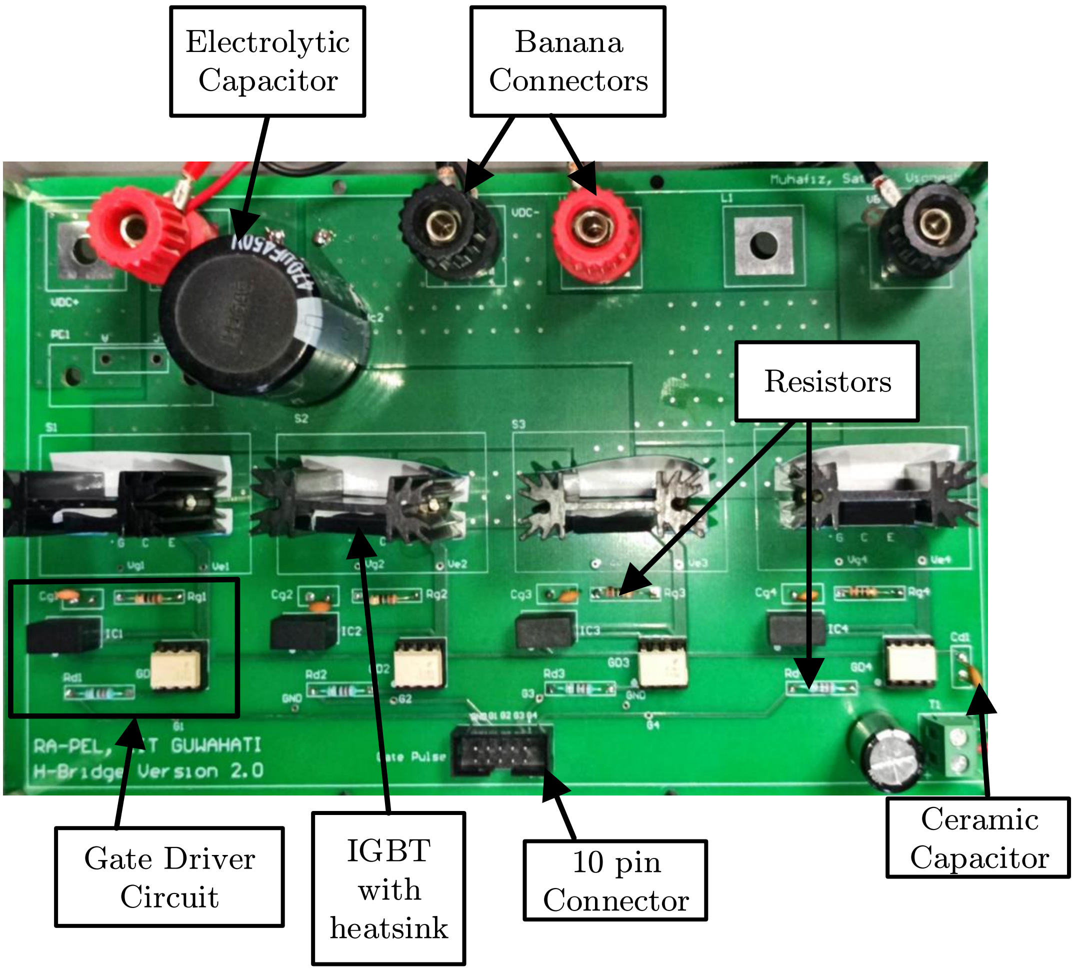


Figure 3.1: H-Bridge Printed Circuit Board Assembly

The PCB was designed in the form of an H-bridge. Each H-bridge consists of four IGBTs.

Selection of switches depends significantly upon the voltage which the switch can handle in OFF condition. According to guidelines, the switch should have at least 1.5 times voltage handling capacity than the operating voltage. In the current case, switches having specifications 1200 V and 60 A were selected for implementing the converter circuit. It may be noted here that in the current scenario, switches need to handle voltages of 48 V and 120 V in their OFF state, on primary and secondary side respectively.

Banana terminals were added to the PCB for supplying input/output to/from the H-bridge circuit. This was intended to connect the power to the circuit and draw its output through these connections. Also, a diode was added in the circuit which could be bypassed using another banana terminal connection. This was done so that the setup could be further extended for application like PV-based systems.

Gate driver circuits were also included in the PCB design. Four gate driver circuits have been implemented for driving four IGBTs in one H-bridge circuit. A voltage of 5 V has been externally supplied to the gate driver circuit which has been increased to 15 V by using isolated DC-DC converter (CRE1S0515SC). The isolated DC-DC converter outputs a regulated 15 V voltage which drives the gate driver IC (FOD3180). FOD3180 has built-in isolators which provide isolation to the gate driver circuit from control circuitry. Block diagram of gate driver circuit is shown in Fig. 3.2. The resistors R_d and R_g were taken as 220 Ω and 15 Ω respectively and C_g was taken as 0.1 μF.

Output of an FPGA board was used to generate switching pulses and these

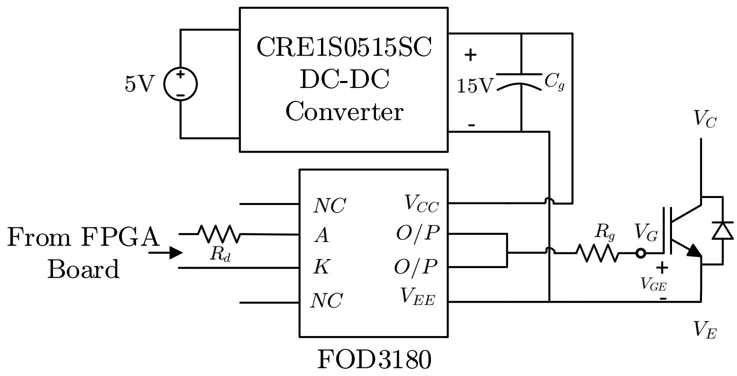


Figure 3.2: Gate Driver Circuit

were given to the gate driver IC. Switching pulses generated were used by gate driver ICs to drive the switches (IGBT).

Although gate driver circuit has been implemented on the same PCB as that of the power circuit, power and gate driver circuits have been segregated in the PCB layout. This was done to make sure that high current flow and switching transient in power circuit section doesn't disturb the operation of gate driver circuit.

3.2.2 Magnetic Component Design

3.2.2.1 Design of High Frequency Transformer

Design of the high-frequency transformer was carried out according to the design process described in [14]. The high-frequency transformer was designed with a power rating of 800 W operating at a switching frequency of 25 kHz. The primary-side voltage was taken to be 100 V, resulting in a rated current of 8 A, while the secondary-side voltage was 240 V.

In a Dual Active Bridge (DAB) converter, the magnetizing current does not transfer any power; thus, it had to be kept small in order to minimize the conduction loss. In this case, the magnetizing current was limited to approximately 5% of the primary current.

Accordingly, using the following relation, the magnetizing inductance was calculated:

$$L_m = \frac{V_1}{4 f_s I_m} \tag{3.1}$$

Based on the above equation, the calculated magnetizing inductance came out to be approximately 2.5 mH.

Once the magnetizing inductance was known, the number of turns on the primary winding was calculated based on the inductance factor of the core used.

$$N_1 = \sqrt{\frac{L_m}{A_L}} \tag{3.2}$$

As per the above equation, the number of turns on the primary side was calculated as 20 turns.

The number of turns on the secondary winding was determined based on the voltage conversion ratio, which can be represented as follows:

$$\frac{N_2}{N_1} = \frac{V_2}{V_1} \tag{3.3}$$

Based on the above relationship and considering the specified voltage levels, the number of turns on the secondary side turned out to be 48 turns.

The essential part of the transformer was an EE100/60/28 ferrite core. Ferrite material (N87) was selected since it had a relatively low core loss at high switching frequencies and therefore would be suitable for operation at tens of kilohertz range. The selected core has an effective cross-sectional area of

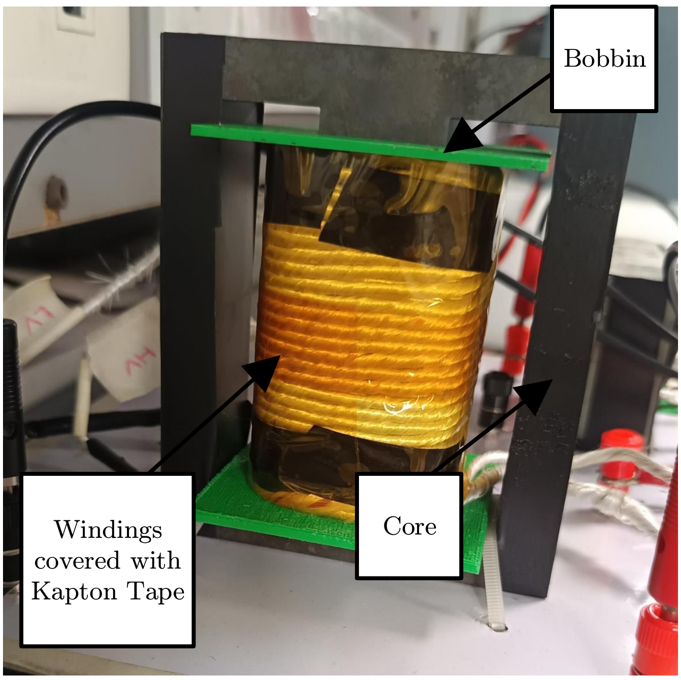


Figure 3.3: High Frequency Transformer Designed for DAB Converter

about 735 mm^2 and an inductance factor (A_L) of 6500–6800 nH/turn^2. It was air-gapless in order to have the highest value of the magnetizing inductance. Conductor dimensions were determined based on the current density. Current density up to 4 A/mm^2 was assumed for safety reasons. Thus, the cross-sectional areas of the conductors were chosen as 2.5 mm^2 and 1.5 mm^2 for the primary and secondary sides, correspondingly. To achieve better magnetic coupling between the windings and to have the lowest value of leakage inductance, both the primary and the secondary windings were put around the center leg of the EE core. Primary winding was located near the core, while secondary was wound after the primary one. These two windings were isolated with kapton tape. Designed transformer is shown in Fig. 3.3. With the above configuration, the calculated magnetizing inductance came out to be approximately 2.5 mH. The fabricated transformer was tested using an LCR meter to determine the magnetizing and leakage inductance. As per the test results, the measured values for the two inductances were 2.8 mH and 5 μH, respectively. Finally, for verification purposes, the designed parameters were cross-checked using Trafolo software.

3.2.2.2 Design of Inductor

Design of series inductor for use in the DAB converter was done by the approach provided in [35]. Series inductor was designed for current ratings of 12 A based on availability of ferrite EE70/33/32 core.

Based on the core selection criterion, which was its availability, design criteria considered were to determine the number of turns as well as the air gap necessary to achieve the required inductance. The number of turns required to achieve the desired inductance was estimated using the following relation:

$$n = \frac{LI_{\max}}{B_{\max}A_c} \times 10^4 \tag{3.4}$$

where n is the number of turns, L is the required inductance value, $I_{\max}$ is the maximum current flowing through the inductor, $B_{\max}$ is the maximum allowable flux density of the core material, and A_c is the effective cross-sectional area of the core.

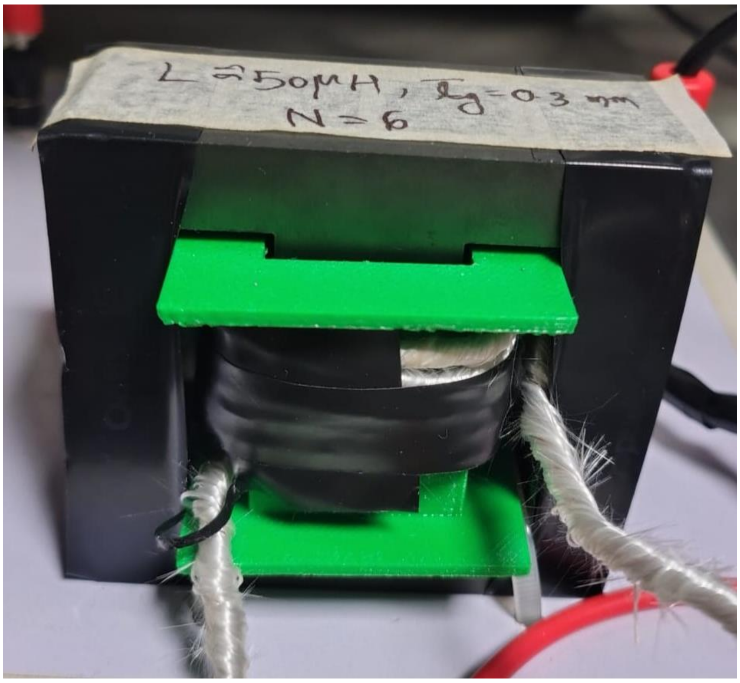


Figure 3.4: Series Inductor Designed for DAB Converter

From equation 3.5, and taking into account the available core parameters, the required number of turns was obtained as 6 turns.

The required air gap required to realize the desired inductance was obtained from:

$$l_g = \frac{\mu_0 A_c n^2}{L} \times 10^{-4} \tag{3.5}$$

where l_g is the air gap and μ_0 is the free space permeability.

Based on the equation 3.6, required air gap for achieving the desired inductance value was about 0.3 mm. The inductor was made using the available core and the number of turns estimated.

Testing of the fabricated inductor was done using LCR meter, and the measured inductance value was about 48.8 μH.

3.2.3 Determining Output Capacitance Value

To obtain a smooth DC voltage at the output of the DAB converter, an output filter capacitor was connected across the load terminals. The value of the output capacitor depends upon the permissible output voltage ripple. In the present case, for an output voltage of 120 V, an allowable peak-to-peak voltage ripple of 0.1% was considered.

Based on the time-domain analysis presented in the Appendix of [36], for the operating condition $m > 1$ and $j_3 < J_{sec} < j_2$, the output capacitance is given by:

$$C_{sec} = \frac{1}{2}\left(\frac{n^2}{(2\pi f_{sw})^2 L \Delta m_{sec}}\right)\left[\frac{\left(\frac{\pi}{2}\left(\delta + (m-1)d_2\right) - J_{sec}\right)^2}{m-1}\right] \tag{3.6}$$

where C_{sec} is the output filter capacitance, n is the transformer turns ratio, f_{sw} is the switching frequency, L is the leakage inductance, Δm_{sec} is the normalized output voltage ripple, m is the voltage conversion ratio parameter, d_2 is the duty parameter of the secondary bridge, δ is the normalized phase shift, and J_{sec} is the normalized average output current.

The normalized quantities are given as:

$$m = \frac{nV_{sec}}{V_{pri}} \tag{3.7}$$

$$\Delta m_{sec} = \frac{n\Delta v_{sec}}{V_{pri}} \tag{3.8}$$

$$J_{sec} = \frac{\pi}{4}(2\delta - \delta^2) \tag{3.9}$$

and the switching instant currents are given by:

$$j_2 = \frac{\pi}{2}\left(\delta + (m-1)d_2\right) \tag{3.10}$$

$$j_3 = \frac{\pi}{2}\left(1 - m + m\delta\right) \tag{3.11}$$

Using the above relations and considering the operating parameters of the converter, the output capacitance was found to be approximately 47 μF. However, considering practical design margin and component availability, a capacitor value of 470 μF was selected for implementation.

3.2.4 Auxiliary Components and Measurement Setup

Alongside the PCB board and magnetic components, some other auxiliary parts were used during the hardware realization of the DAB converter. A DC voltage source was applied for the input supply to the converter.
A regulated supply of 5 V was used as the supply for the gate driver circuit. Two sensors were applied for measuring the input and output voltages of the converter. The signals obtained from the sensors are not suitable for digital processing and had to be transformed by a signal conditioning (sensor) board. The sensor board scaled down the input signals for their further processing by the control hardware. That is, the input high-voltage was

scaled down proportionally with the gain of about 0.004, thus an input of 100 V could be decreased up to 0.4 V.

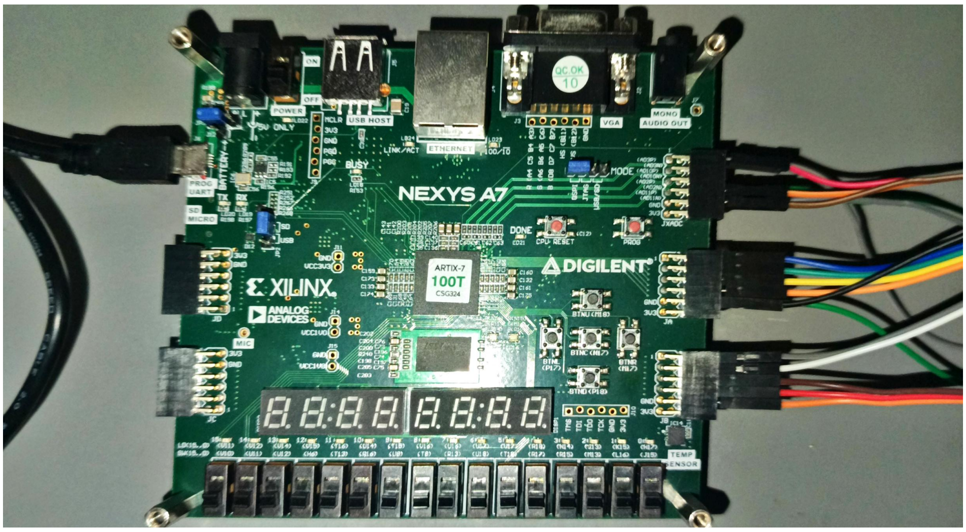


Figure 3.5: Nexys A7 100T FPGA Board

Conditioned signals from the sensor board were provided to the FPGA board (Nexys A7-100T) where control and processing tasks took place. The FPGA board Nexys A7-100T on the basis of the Xilinx Artix-7 structure possesses enough logical resources, fast input-output interface, and good timing control features to perform power electronics-related tasks. The FPGA board allows designing custom control algorithms for digital devices and generating high-frequency switching pulses required for the DAB converter's operation. Besides, it has a parallel processing ability and on-board clock management and communication resources. For measurement purposes, differential voltage probes were used for sensing the input voltage and output (load) voltage. A current probe was used for sensing inductor current on the transformer's primary winding. Proper isolation and high frequency signal sensing capabilities of the probes allowed accurately measuring high-frequency signals without interference. All the measured waveforms were recorded using a digital storage oscilloscope (DSO). Additionally, a resistive load was connected to the output of the converter. The complete

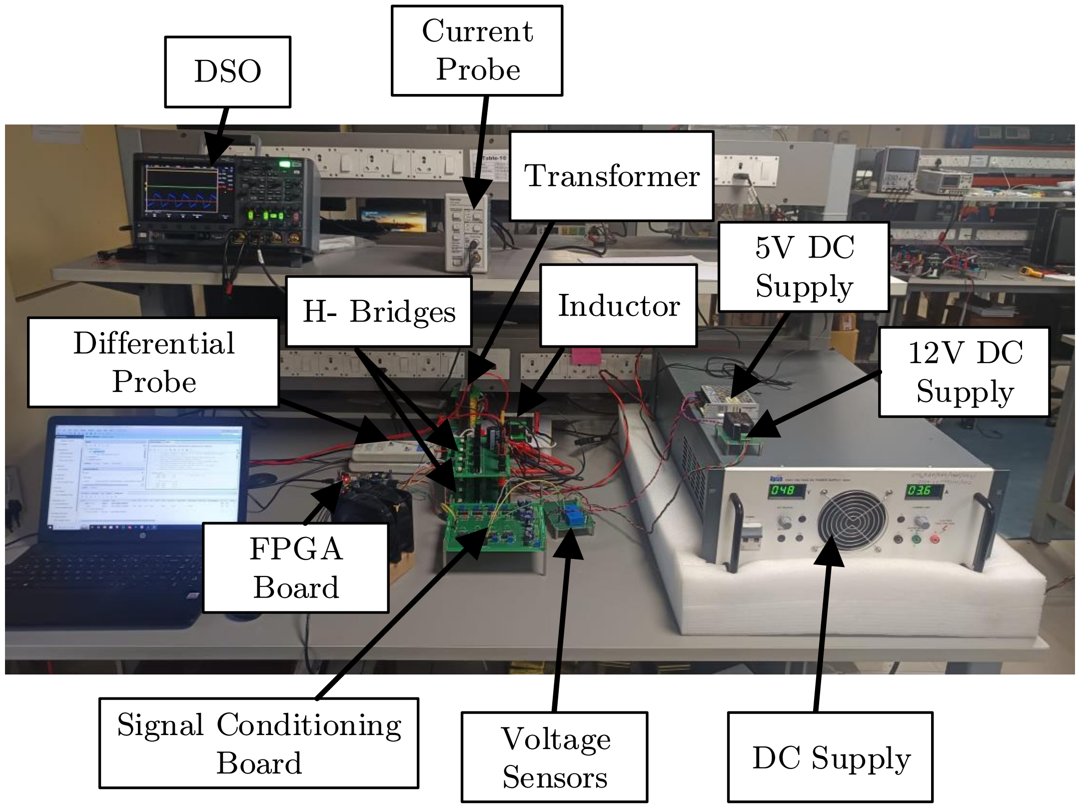


Figure 3.6: Hardware Setup of DAB Converter

hardware setup of DAB converter is shown in Fig. 3.6.

3.3 Validation of Open-Loop DAB Operation

3.3.1 Hardware Parameters

Table 3.1: Hardware parameters of the DAB converter

Parameter	**Value**
Input DC Voltage, V_{dc1}	48 V
Output Voltage, V_{dc2}	120 V
Switching Frequency, f_{sw}	25 kHz
Load Resistance, R_L	100 Ω
Transformer Turns Ratio, N	20/48
Transformer Leakage Inductance, L_r	5 μH
Series Inductance, Ls	50 μH
Rated Power, P_{rated}	144 W

Applying the power transfer formula (Eq. 2.1) and plugging in the values in Table 3.1, the angle of phase shift that would be required at the secondary bridge was computed. At an operational value of 144 W, it was determined that the angle of phase shift would be around $37.52°$. The phase shift implies a time difference of approximately 4.17 μs.

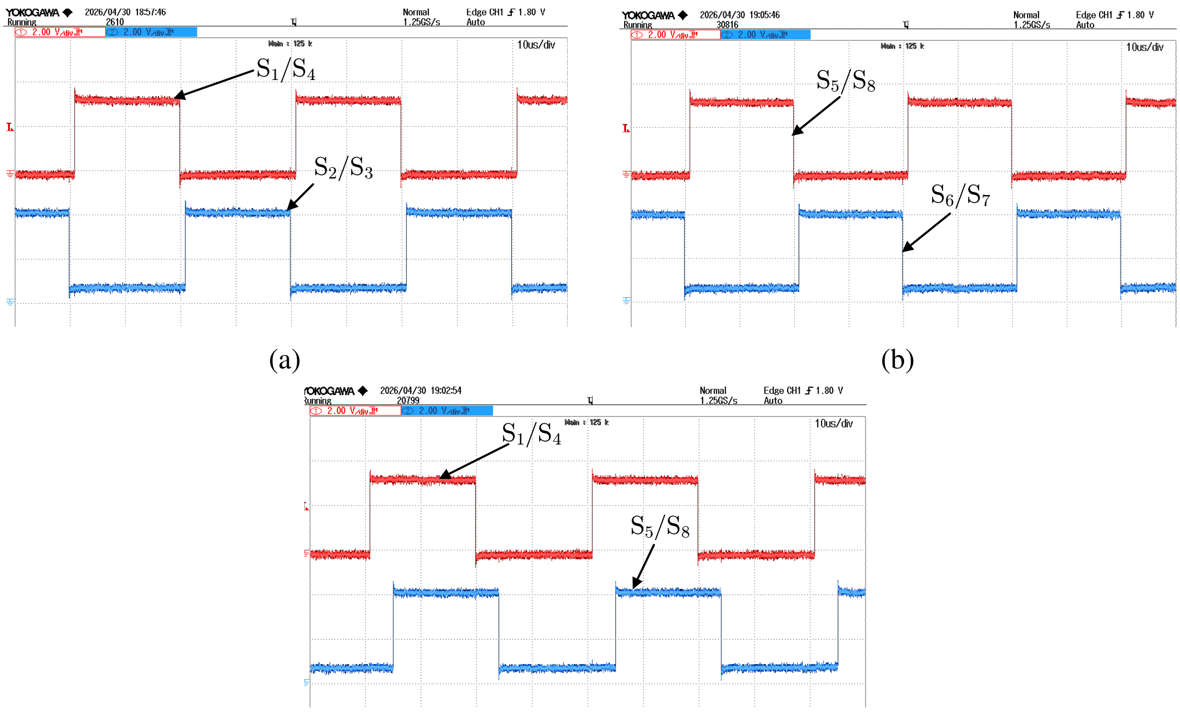
3.3.2 Generation of Switching Pulses

For the hardware implementation, the switching pulses were generated in the same manner as described in Section 2.1, using a square waveform with a 50% duty cycle and a phase shift between the primary and secondary bridges. However, in practical hardware, non-ideal characteristics of switching devices must be considered. In particular, the IGBTs used in the converter require a finite turn-off time before the complementary switches can be safely turned on. If this delay is not properly maintained, both switches in the same leg may conduct simultaneously, resulting in a shoot-through condition that can damage the converter.

The generation of the switching pulses has been implemented using a digital design on the FPGA (Nexys A7 100T). Firstly, a high frequency triangular waveform of a carrier was created using an up-down counter. Then, the reference value is compared with this waveform to get the raw PWM signal with duty cycle ratio 50%.

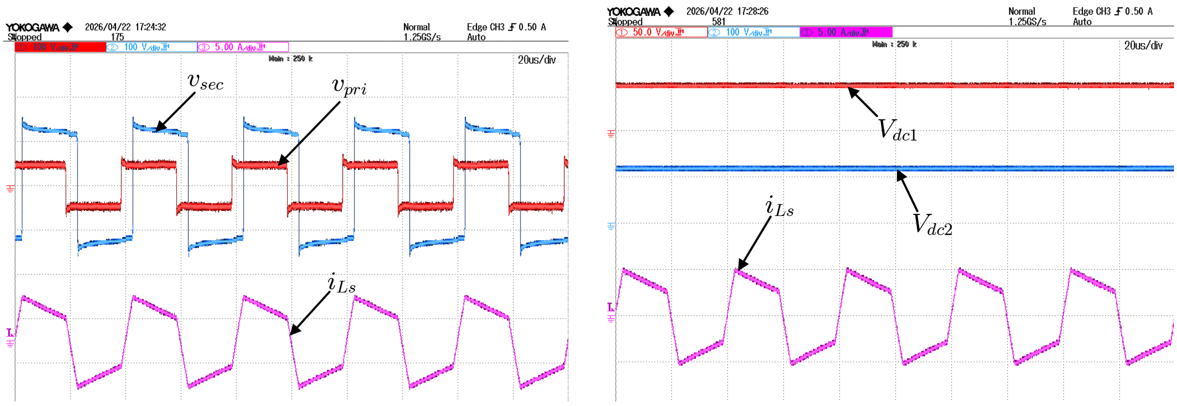
A delay line was implemented using a shift register block to implement the concept of dead time. A fixed dead time of 1 μs was kept before turning ON the next set of switches to prevent any shoot-through conditions.

3.3.3 Hardware Results



(a) (b) (c)

Figure 3.7: Switching Pulses Generated Using Nexys A7 100T in Open-Loop Hardware Implementation



(a) (b)

Figure 3.8: Open Loop Hardware Results

3.3.4 Observation and Discussion

The switching pulses produced for the converter are depicted in Fig. 3.7, whereas the related waveforms of the hardware are displayed in Fig. 3.8. In the experiment, upon providing an input voltage of 48 V, it was noticed that the input current value is approximately 3.3 A. The obtained output voltage value through hardware is about 119.1 V, which is relatively close to the theoretical value.

It is noticed that the peak value of the inductor current obtained from the experiment is approximately 5 A, which is relatively greater than the value obtained through simulation (around 4 A). The increment in the current value could be due to some practical non-idealities like switching noise and other parasitics associated with the hardware circuitry.

The efficiency of the converter was found to be approximately 90%. The decrement in efficiency value can be associated with the switching losses of the IGBTs, especially the turn-off losses and other non-idealities existing within the hardware circuit.

3.4 Validation of Closed-Loop DAB Operation

3.4.0.1 Hardware Parameters

The hardware parameters that were used for closed-loop experimental validations were identical to those mentioned earlier in Table 3.1. The input voltage, switching frequency, transformer parameters, and series inductance remained unchanged in comparison with open-loop experiments. Nevertheless, the controller robustness was investigated by varying load resistance during the process. This was done by changing load resistance from 100 Ω to 200 Ω. Such a change made it possible to verify the effectiveness of the controller to regulate the output voltage in the presence of changes in the load resistance.

3.4.1 Implementation of Closed Loop Control

The closed loop control algorithm implementation followed the same principles as those described previously in the simulation studies. Specifically, this study used the Xilinx Artix-7 based FPGA development board called Nexys A7-100T.

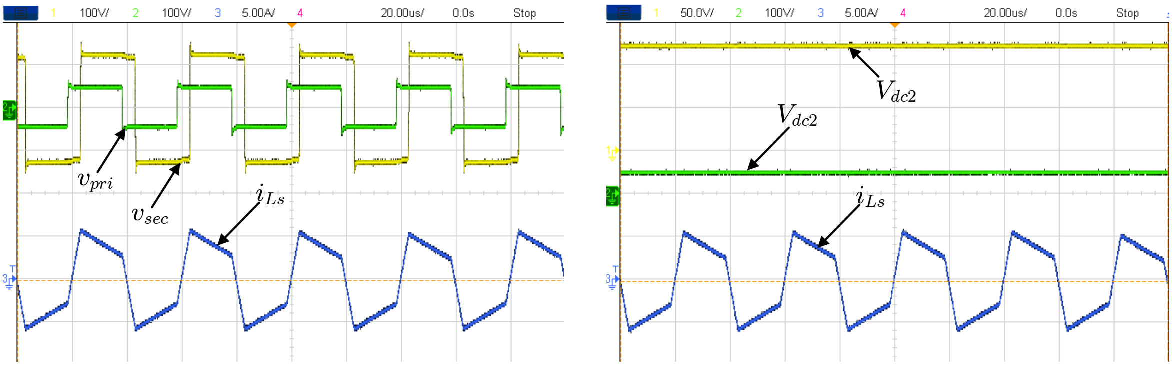
For sensing the output and input voltage of the converter, two voltage sensors were employed, and the obtained voltage signals were sent to a signal conditioning (or sensor) board. In particular, the voltages were scaled by about 0.004 (1/250) to bring the high voltage level to a value compatible with the FPGA ADC operation, i.e., within the range 0-1 V. Then, these voltage signals were applied to the ADC channels of the FPGA board, and the samples of input and output voltages were acquired. The voltage signals were sampled at the peak of the triangular wave generated for the pulse generation in the converters. To make sure the voltages were sampled accurately, a small delay of 1.5 μs was applied. The reference voltage was equal to $V_{\text{ref}} = 2.5V_{\text{in}}$. An error voltage was computed as a difference between the reference and measured voltage, and then this voltage was used as the input for the PI controller. The output of the PI controller gave an indication for the required phase shift of the secondary bridge switching pulses. The PI controller execution was repeated only once per triangular carrier period (once per switching cycle).

3.4.2 Generation of Pulse Waveform

For pulse generation in the proposed method, the FPGA board was used after designing the closed loop control algorithm. For generating the pulses in the closed loop method, SPS technique was chosen here. According to the SPS technique, there is switching at a 50% duty cycle of both primary and secondary bridges. The transfer of power through the bridges could be controlled by phase shift among these two bridges.

The switching pulses of the primary bridge were steady while those of the secondary bridge were phase shifted in relation to the pulses of the primary bridge. The appropriate value of phase shift could be determined from the output of the PI controller. The PI controller was continuously varying the phase shift for obtaining the desirable voltage level. The switching pulses could be obtained with the use of the carrier frequency signal in FPGA. Phase shift values were being updated continuously so that the accurate switching pulses can be obtained. Some dead time was also introduced between complementary pulses to prevent any possibility of shoot-through currents. Furthermore, the synchronization of the pulses of primary and secondary bridges was made to ensure appropriate working of the converter. This ensures the stability and glitch-free signals. In addition, continuous variation of the phase shift helps in achieving power transfer smoothly as well as controlling the output voltage.

3.4.3 Hardware Results



(a) (b)

Figure 3.9: Closed-Loop Hardware Results

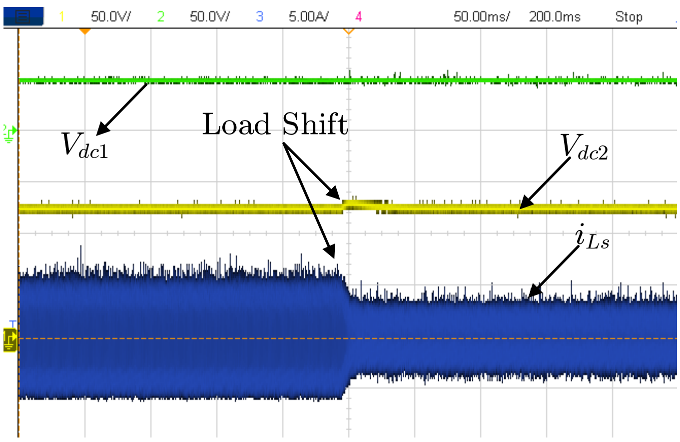


Figure 3.10: Transients in Load Voltage and Inductor Current after Load Switch

3.4.3.1 Observation and Discussion

In this case, the voltage regulator operates in closed loop with the input voltage of about 48 V. In steady state, the output voltage of the device is found to be in the range of 120–123 V. For small variations of the input voltage from 48 V up to 48.9 V, the output voltage is maintained to be regulated near this value which represents twice-and-a-half the input voltage, according to the reference value. The performance of the system in terms of dynamics was tested when the load resistance varied from 100 Ω up to 200 Ω as shown if Fig. 3.10. During this process, the system exhibits the transient response, which lasts about 40 ms. Following the transient stage, the output voltage stabilizes at the level of 120–122 V.

From these observations, it can be concluded that using the closed-loop control system, the proper voltage regulation can still be provided under changing load conditions. The ability to recover from the disturbance indicates the efficiency of the applied control method.

In general, the use of the closed-loop control system guarantees the required level of stability and voltage regulation compared to the open-loop system.

Chapter 4

Conclusion and Future Work

In this project, the design of a power factor corrected converter along with the simulation and hardware implementation of a dual active bridge (DAB) converter was done. Initially, an overview of the rise in usage of electric vehicles in India and the corresponding demand for efficient charging facilities was provided.

As a part of the study, a converter with two stages, viz., an AC-DC front-end followed by an isolated DC-DC DAB converter was designed. Unit vector template (UVT)-based control was used for the front-end stage to achieve near-unity power factor with the help of power factor correction. From the simulation results, it was found that UVT-based PFC could successfully maintain the power factor close to unity and improve input current quality. Then, DAB converter was analyzed in both open-loop and closed-loop modes. For the closed-loop mode, the relationship between power and phase shift in the converter was identified. Further, the required control method was designed to regulate the output voltage at a certain value. Simulation results showed good agreement with theoretical considerations with respect to phase shift-power transfer characteristics and closed-loop regulation of output voltage.

Moreover, hardware implementation of the DAB converter was done. An IGBT-based H-bridge, high frequency transformer, and FPGA-based gate driver circuit were used to implement the converter. Experimental results confirmed the proper operation of the converter. The output voltage was close to the desired value and acceptable efficiency was achieved. From the experimental results, it can be concluded that satisfactory agreement was achieved between simulation and hardware implementations of the DAB converter. However, some deviations between simulations and experimental results are expected due to the effect of practical losses such as switching losses.

Thus, this study proved that combining power factor correction at the front-end with isolated DC-DC conversion by using DAB is a suitable choice for EV chargers.

There are many aspects of this work that can be improved in the future for better performance and scalability.

Firstly, the power density of the converter needs to be increased by using a smaller high-frequency transformer. To do this, switching frequency should be increased significantly by using wide bandgap semiconductors such as Silicon Carbide (SiC) or Gallium Nitride (GaN). Secondly, the power rating of the converter can be increased by using SiC-based devices and improved cooling systems to ensure reliable operation at higher frequencies.

Furthermore, advanced control techniques such as dual phase shift (DPS), extended phase shift (EPS), etc. can be adopted to improve efficiency, reduce current stresses on devices and maximize soft switching region. Moreover, other isolated converters such as triple active bridge (TAB) can be analyzed in order to explore their potential in multiport power conversions and energy storages. Moreover, the converter can be extended to include battery charger algorithms, feedback control using FPGA or digital signal controllers. Thus, the present work forms a solid basis for future research into high efficiency converters for next-generation electric vehicle chargers.

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